Colour evidence for a same-redshift quasar companion: the unified 41-band model (October 2026)

Internal method note, `qso_pcolor`

4 October 2026

Abstract

A quasar has a spectroscopic redshift z_0, and a point source lies a few arcseconds away. This note describes how `qso_pcolor` measures, from broad-band photometry alone, how strongly the companion's colours support the hypothesis that it is itself a quasar at z_0. The method compares four populations: quasars at z_0, quasars at other redshifts, everything else in the imaging catalogue (the background, mostly stars), and a broad catch-all. It accepts any subset of 41 bands from seven surveys that includes Legacy Surveys r at signal-to-noise ≥ 10; without Legacy r, a fallback band stands in and Legacy r is integrated over (Section 12). One latent model, shared by the northern and southern Legacy Surveys, describes all bands, and photometry is corrected for Galactic extinction. Two complete models are provided, and both are meant to be used. They share everything except the quasar colour model: in one, quasar colours may depend on apparent magnitude (`current`); in the other they do not, the XDQSO assumption that protects against learning the training sample's selection (`current_magindep`). The background is fitted, as in XDQSO, separately in eight bins of Legacy r, on a training sample from which likely quasars are removed, with component weights that vary smoothly with Galactic position. The abundance of quasars at each magnitude and redshift is the completeness-corrected eBOSS quasar luminosity function. Objects whose colours are unlike both models are set aside by a test calibrated per object for its own bands and noise. On held-out data both models separate quasars from background objects with AUC 0.993–0.994, and 0.6% of quasars are left unranked.

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1 Summary of what is delivered

- **Two modes.** *Resolved mode* (`UnifiedPSFModel`): the companion is measured as its own source, and the question is whether it is a quasar at z_0 (Sections 3–10). *Blend mode* (`BlendModel`): the quasar and companion are measured as one source, and the question is whether the hidden companion is a second quasar at z_0 or a star (Section 11).

- **Two models, both for use** (promoted 4 October 2026): `models/multisurvey_psf/current`, with magnitude-dependent quasar colours, and `models/multisurvey_psf/current_magindep`, with magnitude-independent quasar colours (the XDQSO assumption). `current` names the magnitude-dependent model, which recovers more test quasars at $p_{\rm quasar} > 0.5$, but the magnitude-independent model can and should be used alongside it: the two rank equally well overall, and it guards against a risk our tests cannot measure (Section 6.2). Score every candidate with both. Load with `UnifiedPSFModel.load(...)`.

- **Inputs.** Any subset of 41 bands (SDSS *ugriz*; Legacy Surveys DR9 South and North $g, r, z, W1, W2$; AllWISE $W1$–$W4$; Pan-STARRS1 *grizy*; NSC $ugrizY{+}VR$; SkyMapper *uvgriz*; VHS $YJHK_s$) that contains Legacy r at $S/N \geq 10$, as catalogued fluxes or magnitudes with errors; candidate RA and Dec; the primary's redshift z_0; morphology and blend information. Fainter sources are flagged and not scored (Section 4.1); sources without Legacy r are scored, by default and with a warning, by integrating over Legacy r given a fallback band (Section 12).

- **Output.** A ranking statistic $\log \mathcal{R}$ (Section 3), evidence and posterior quantities, a quasar-support percentile, and explicit eligibility and quality flags.

- **Main results** (Section 10; first number for the magnitude-independent model, second for the magnitude-dependent one). AUC 0.993 and 0.994 on held-out quasars and background objects; 91% and 94% of quasars receive $p_{\rm quasar} > 0.5$ (at $23 < r < 24$, 30% and 54%); 0.33% and 0.44% of background objects that are not flagged as likely quasars do, most plausibly unlabelled faint quasars (Section 9.1). 0.6% of quasars are not ranked, with no dependence on the number of observed bands (Section 9.2). No artificial colour-grid point outside the models' dense core receives $p_{\rm quasar} > 0.5$. Among ranked objects $p_{\rm quasar}$ agrees with the observed quasar fraction where the labels are complete; its faint end cannot be tested with the available labels, and above 0.97 it exceeds the known-quasar fraction mostly because the background sample holds unlabelled quasars. 1.0–1.2% of known quasars in a natural sample are not ranked.

2 Notation and glossary

2.1 Symbols

Symbol	Meaning (units)
z_0	spectroscopic redshift of the primary quasar
z	redshift of a hypothetical companion quasar
$\boldsymbol{y}$	observed native luptitudes of a source in the bands it has (mag; Section 5)
$\boldsymbol{C}$	measurement covariance of $\boldsymbol{y}$ (mag^2; diagonal in practice)
$\boldsymbol{x}$	latent, noise-free luptitudes in the 36 shared coordinates (mag)
$H,\ \boldsymbol{b},\ T$	observation operator: $\boldsymbol{y} = H\boldsymbol{x}+\boldsymbol{b}+\boldsymbol{\epsilon}$, with extra instrumental scatter $\boldsymbol{\epsilon} \sim N(0, T)$
m_{ref}	luptitude of the reference band, Legacy r, on which colours are conditioned (mag)
a	index of the reference band
E	SFD98 reddening $E(B-V)$ at the source position (mag)
R_j	extinction coefficient of band j, A_j/E
$\ell,\ b$	Galactic longitude and latitude (deg)
$\Sigma_Q(z, m)$	quasar surface density per unit redshift and magnitude (deg^{-2} mag^{-1})
$\Sigma_B(m)$	background surface density per unit magnitude (deg^{-2} mag^{-1})
η	catch-all share of the background intensity (dimensionless)
λ_X	intensity of hypothesis X: surface density times colour likelihood
$\mathcal{R}$	ranking statistic, Eq. (3) (per unit redshift)
Δz_{eff}	effective width of the declared redshift window, `dz_match_eff`
K	number of Gaussian components in a mixture
$w,\ \nu$	covariance-prior scale (10^{-3} mag^2) and strength (1)

2.2 Terms

Luptitude. The inverse hyperbolic sine magnitude of Lupton et al. (1999),

$$m = 22.5 - \frac{2.5}{\ln 10}\left[\sinh^{-1}\left(\frac{f}{2s}\right) + \ln s\right],$$

for flux f in the band's native unit and softening s. It equals the ordinary magnitude for $f \gg s$ and stays finite for zero or negative flux.

Native system. A band's own catalogue calibration. No band is converted into another survey's system.

Slice. The quasar model for redshifts within ± 0.1 of a centre z_k, for $z_k = 0.15, 0.25, \ldots, 4.35$ (43 slices; neighbouring slices overlap).

Reference band. Legacy DR9 r: the southern band where observed, otherwise the northern one. Colour likelihoods are conditioned on it, and the background bins are defined in it. A scored source has it at $S/N \geq 10$ (Section 4.1), or has it integrated over (Section 12).

Resolved mode, blend mode. The two ways the package is used. Resolved mode scores a companion measured as its own source (quasar at z_0, quasar at another redshift, background, catch-all). Blend mode scores a single source that is the quasar plus an unresolved companion (QQ against QS; Section 11).

Fallback band. For a source without Legacy r: the band (PS1 r, NSC r or SkyMapper r) that sets eligibility and centres the integral over the unknown Legacy r (Section 12).

Role. Every training object belongs to one sky-based role: *fit* and *select* (training), *calibration* (catch-all shares and support threshold), or *test* (validation). Roles are fixed in advance and never reassigned.

Background. Every point source in the imaging catalogue that is not a quasar. It is dominated by stars and is fitted empirically. Its training sample excludes known quasars and likely quasars (Quaia members and WISE AGN colours; Section 7).

Catch-all. A broad Student-t density with a small share of the background intensity. It explains objects that neither the quasar nor the background mixture fits, so that Gaussian tails do not decide their classification.

Support percentile. For a candidate and z_0, the fraction of simulated quasars at z_0, with the candidate's own errors, whose conditional density is lower than the candidate's. A small value means the colours are unusual for a quasar at z_0.

Outside both models. An object whose colours are unlike both the quasar model (at any redshift) and the background model. It is not ranked. Since 5 October 2026 the decision uses calibrated tail probabilities p_Q, p_B of the nearest-component distance (Section 9.2), not a fixed number of σ.

Held-out density. The mean, over test objects, of the conditional log density $\ln p(\boldsymbol{y}_{\rm other} \mid m_{\rm ref})$ (nat per object, density per mag^{N-1}). Higher is better.

AUC. The probability that a random test quasar ranks above a random test background object on $\log \mathcal{R}$. Objects the scorer declines to rank are placed at the bottom.

3 The question and the ranking statistic

Four hypotheses compete for a companion with photometry $\boldsymbol{y}$ near a primary at z_0 (Fig. 1):

1. $Q(z_0)$, a quasar at the primary's redshift, within a declared window;
2. $Q(z \neq z_0)$, a quasar at any other redshift;
3. B, a member of the background population;
4. U, the catch-all.

Each hypothesis has an intensity, a surface density times a likelihood. With reference magnitude m and conditional colour likelihoods,

$$\lambda_Q(z) = \Sigma_Q(z, m)\, p(\boldsymbol{y} \mid Q, z, m), \tag{1}$$

$$\lambda_B = (1-\eta)\, \Sigma_B(m)\, p(\boldsymbol{y} \mid B, m, \ell, b), \qquad \lambda_U = \eta\, \Sigma_B(m)\, p(\boldsymbol{y} \mid U, m). \tag{2}$$

We rank candidates on

$$\mathcal{R} = \frac{\lambda_Q(z_0)}{\int \lambda_Q(z)\, dz + \lambda_B + \lambda_U} \qquad \text{[per unit redshift]}. \tag{3}$$

The denominator does not depend on the redshift window, because same-redshift and other-redshift quasars together form the whole quasar intensity. A posterior for a window of effective width $\Delta z_{\rm eff}$ follows by one multiplication, $p_{\rm same} = \mathcal{R}\, \Delta z_{\rm eff}$ (exact for the window-averaged $\mathcal{R}$ that the scorer stores). Photometric redshifts have widths of about $\sigma_z \approx 0.6$, while a $\pm 2000\,{\rm km\,s^{-1}}$ window at $z = 1.4$ is $\Delta z = 0.016$ wide, so $p_{\rm same}$ is small even for an ideal candidate. $\log \mathcal{R}$ is the quantity to compare between candidates. Appendix A gives the derivation.

4 Data and roles

Quasars. The training quasars are the union of DESI DR1 quasars (`spectype=QSO`, `zwarn=0`, primary) and SDSS DR16Q (Lyke et al., 2020), merged by sky position. Of 1,657,444 eligible quasars, 1,049,260 are fit or select rows. Of these, 610,503 are DESI-only, 241,808 SDSS-only, and 196,949 in both (Fig. 4). The remaining quasars are 239,371 calibration and 368,813 test rows. Their photometry was acquired from all seven surveys and stored as native fluxes and variances.

Background. The background sample is 2,978,857 cleaned point sources in 274 sky regions (Legacy `type=PSF`, mask-clean, known quasars removed), with seven-survey photometry. Of these, 1,975,894 are fit or select rows and 502,251 calibration rows.

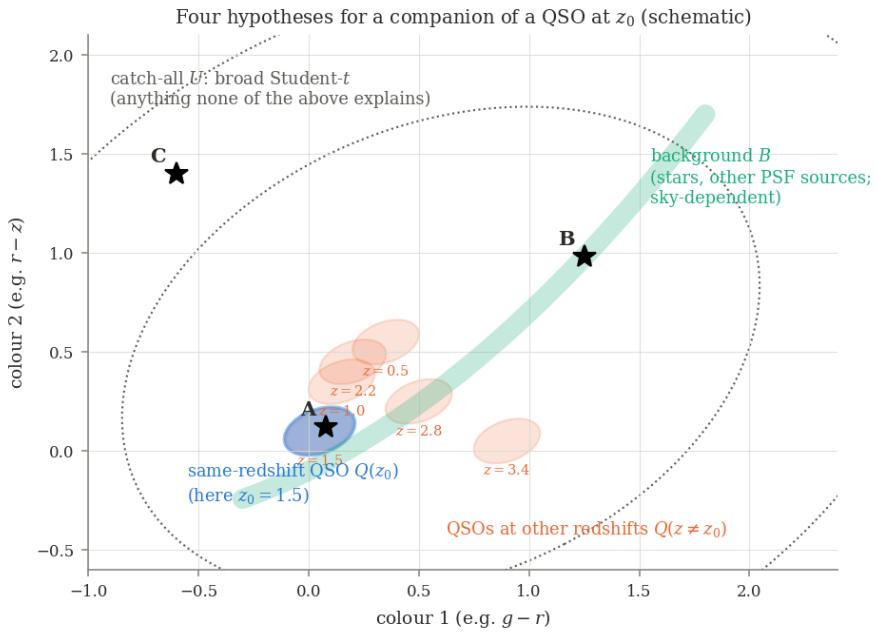


Figure 1: Schematic (synthetic, not fitted) of the four hypotheses in one colour plane. Candidate A lies on the same-redshift quasar locus, B on the background locus, and C far from both; the catch-all explains C, so the Gaussian tail of whichever population is wider does not decide it.

Selection. All samples satisfy $|b| \geq 25°$, and both modes refuse sources below it. The limit marks where the models were trained and checked, not a physical boundary: nearer the plane the stellar density rises (companions are more often stars, crowding and blending more common), the stellar mix changes (thin disc, and bulge towards the centre), and extinction becomes large and patchy, where the SFD98 correction is least reliable; the spectroscopic quasar samples also barely cover the plane. A locally refitted background model was compared with the global one in 30 cones at $|b| = 20$–$78°$: at 20–45° it fitted the field better by 0.1–0.2 nat per object, with unchanged false-positive rates. Nothing below 20° has been tested. A band is used only when its survey marks it usable (finite flux, positive variance, a positive exposure count where reported, and the survey's cleanliness flag); weak and negative fluxes are kept. The quasar models were trained without a signal-to-noise cut (0.7% of their fit/select rows lie below the faint limit). The background is trained, and all objects are scored, only above the faint limit of Section 4.1.

Roles. Roles are assigned by sky region before any fitting (Fig. 3). The background regions are 274 cones of radius 0.3°: 146 fit, 34 select, 45 calibration and 49 test cones, so test objects lie in separate sky regions. Rows used during development to choose stopping points or thresholds (34,259 quasars and 2,045 background objects for the quasar fits; the stopping and evaluation rows of the background design tests) are recorded and excluded from the tests in Section 10.

4.1 The faint limit

We train the background on, and score, only sources with Legacy r at $S/N \geq 10$ ($\sigma_r \lesssim 0.11$ mag; $\sigma(g-r) \approx 0.2$ mag). Below this the colour errors are as large as the colour difference between quasars and stars, and the model cannot vouch for a quasar. Measured before the limit was imposed (same quasar models, joint background), the fraction of held-out quasars given $p_{\mathrm{quasar}} > 0.5$ is 19% at $5 \leq S/N < 7$ and 30% at $7 \leq S/N < 10$, against 51% at $10 \leq S/N < 15$ and 93% above $S/N = 50$ (Fig. 2). The limit is a signal-to-noise cut rather than a magnitude cut because the two hemispheres differ in depth: stellar counts peak at $r \approx 24.6$ in the South and $r \approx 23.6$ in the North, and $S/N = 10$ falls at $r \approx 24.1$ and 22.9 respectively. It removes 2.8% of test quasars and 64% of test background objects, almost all faint stars. Legacy r is then the reference band of every

scored source, so the quasar and background densities are conditioned on the same magnitude that defines the background bins. A source without Legacy r (for example outside the Legacy footprint) is scored by integrating over its unknown Legacy r given a fallback band (Section 12).

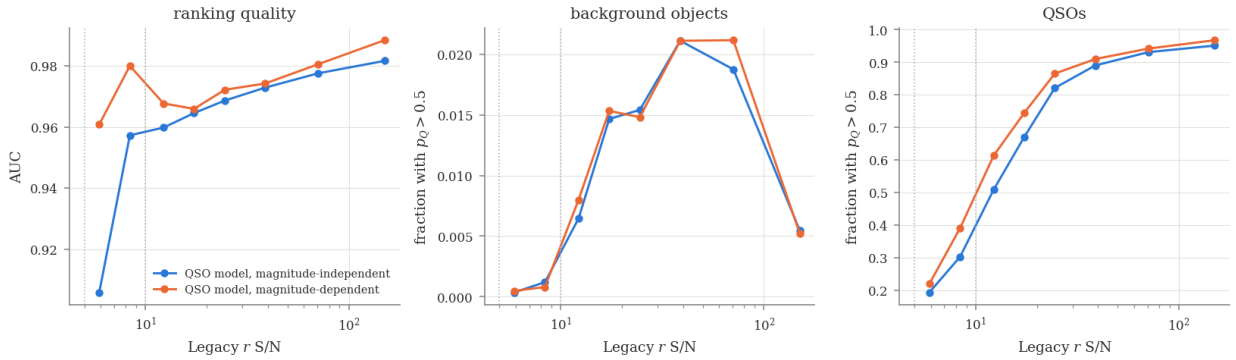


Figure 2: Held-out performance against Legacy r signal-to-noise, measured before the faint limit was imposed. Left: AUC within each S/N bin. Middle and right: fraction of background objects and of quasars given $p_{\text{quasar}} > 0.5$. Dotted lines: $S/N = 5$ and 10. Scored with the earlier fixed 4σ outside-both test, which chose the faint limit; Section 9.2 describes the test now used.

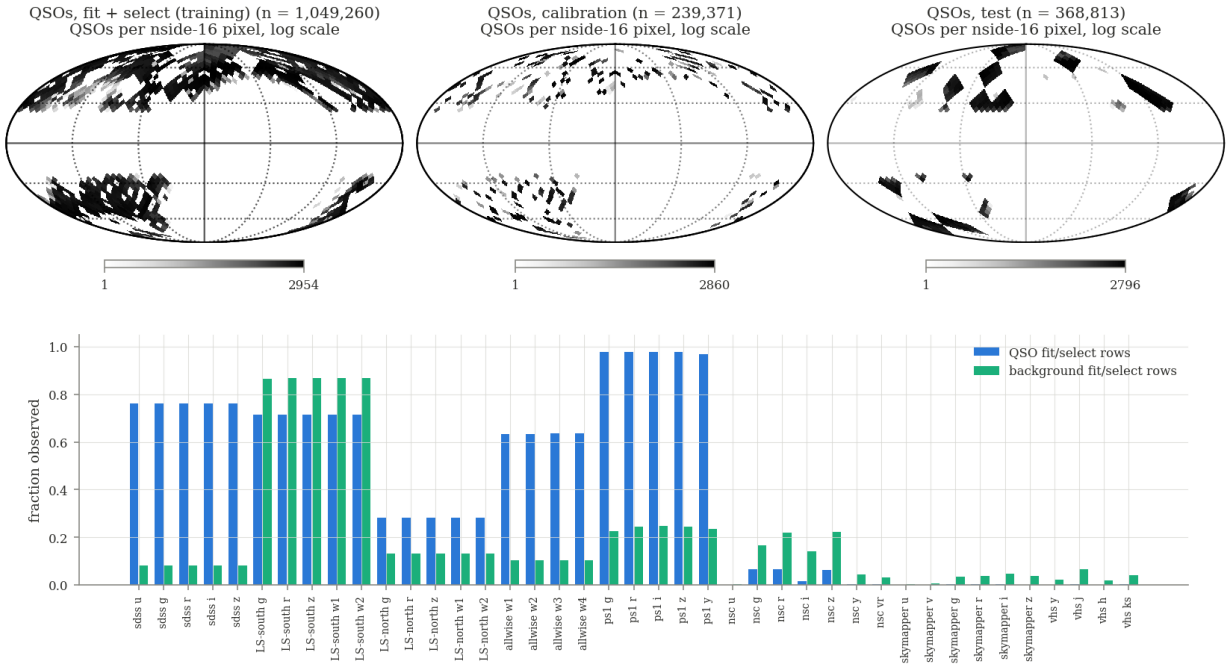


Figure 3: Top: sky distribution of quasars by role, Galactic Mollweide projection, counts per HEALPix nside-16 pixel (log scale). Bottom: fraction of fit/select rows with each band observed, for quasars (blue) and the background (aqua).

5 Photometry and the observation model

5.1 Native luptitudes and measurement errors

Every band keeps its native calibration and zero point (AB for SDSS, Legacy, PS1, NSC and SkyMapper; Vega for AllWISE and VHS). Fluxes become luptitudes with a per-band softening s, and their variances propagate through the exact derivative of the luptitude function. A band that is not observed is marginalised exactly; it is never imputed.

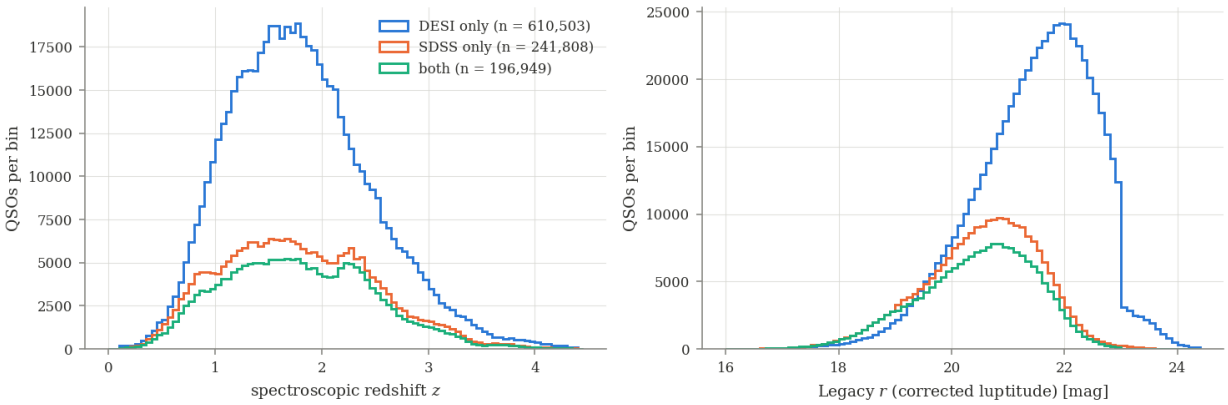


Figure 4: Fit/select quasars by parent survey. Left: spectroscopic redshift. Right: Legacy r (extinction-corrected luptitude, southern band where observed, otherwise northern). The DESI-only sample ends near DESI's quasar target limit, $r \approx 23$.

5.2 Galactic extinction

Fluxes and variances are corrected before any other step:

$$f \to f\,10^{0.4R_jE}, \qquad \sigma_f^2 \to \sigma_f^2\,10^{0.8R_jE}, \tag{4}$$

where E is the SFD98 reddening (Schlegel et al., 1998) at the source position. This preserves signal-to-noise and keeps negative fluxes negative. Table 1 lists the coefficients. Legacy DR9 values come from the survey documentation (Legacy Surveys, 2020), SDSS and PS1 values from Schlafly & Finkbeiner (2011), Table 6, and SkyMapper values from Wolf et al. (2018). NSC VR and VHS Y have no published values in these sources. We computed them with the Schlafly & Finkbeiner (2011) method: an F99 law with $R_V = 3.1$ and a 7000 K source, with one scale factor fitted to 16 published rows (residual rms 0.73%). Top-hat bandpasses were used, so these two values are approximate; we estimate ±5% for VR. The scorer computes E from the candidate's position and applies the same correction, so users supply catalogue fluxes as observed. Fig. 5 shows that our correction reproduces the Legacy catalogue correction (`mw_transmission`) to 0.002 mmag.

Table 1: Extinction coefficients $R_j = A_j/E(B-V)_{\rm SFD}$.

Bands	R_j	Source
Legacy DECam g, r, z	3.214, 2.165, 1.211	Legacy DR9 documentation
Legacy North g, r, z	3.278, 2.189, 1.196	DECam values through the DESI relation, Eq. (6)
$W1, W2$ (Legacy, AllWISE); $W3, W4$	0.184, 0.113; 0.0241, 0.00910	Legacy DR9 documentation
SDSS *ugriz*	4.239, 3.303, 2.285, 1.698, 1.263	Schlafly & Finkbeiner (2011), $R_V = 3.1$
PS1 *grizy*	3.172, 2.271, 1.682, 1.322, 1.087	Schlafly & Finkbeiner (2011)
NSC *ugrizY*	3.995, 3.214, 2.165, 1.592, 1.211, 1.064	DECam (Legacy DR9)
NSC VR	2.28	computed, approximate
SkyMapper *uvgriz*	4.294, 4.026, 2.986, 2.288, 1.588, 1.206	Wolf et al. (2018)
VHS J, H, K_s; Y	0.709, 0.449, 0.302; 1.00	UKIRT rows of Schlafly & Finkbeiner (2011); Y computed

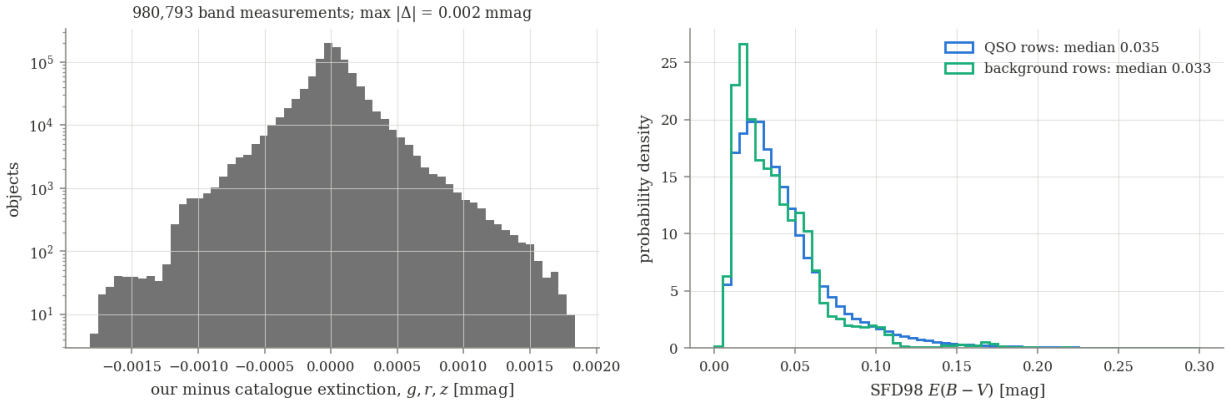


Figure 5: Left: our correction minus the Legacy catalogue correction, in four reserved Legacy cones (980,793 g, r, z measurements). Right: distribution of SFD98 reddening for the quasar and background rows (medians 0.035 and 0.033 mag).

5.3 One latent model for Legacy North and South

Legacy DR9 South (DECam) and North (BASS/MzLS) g, r, z use different filters. So that the northern fits can borrow from the much larger southern sample, we describe every source by latent, noise-free luptitudes $\boldsymbol{x}$ in 36 coordinates: five shared Legacy coordinates (`legacy:g,r,z,W1,W2`) and 31 coordinates for the other surveys. The 41 native bands follow from a linear operator,

$$\boldsymbol{y} = H\boldsymbol{x} + \boldsymbol{b} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim N(0, T). \tag{5}$$

Rows of H for the southern Legacy and all non-Legacy bands are unit vectors. Rows for the northern g, r, z carry the DESI North→South colour relation. For example, $y_{g,N} = 1.0627\, x_g - 0.0643\, x_r + 0.0016\, x_z - 0.0060$. T holds the instrumental scatter measured on overlap pairs. The relation acts on uncorrected magnitudes, so extinction is removed in the DECam system:

$$R_{\text{native}} = H\, R_{\text{latent}}, \tag{6}$$

where each latent coordinate takes the coefficient of the system it represents. Fig. 6 shows that the relation removes the colour-dependent North–South offsets on the same objects. The Legacy $W1, W2$ coordinates are shared after both hemispheres use common units and softening; they remain distinct from AllWISE $W1, W2$.

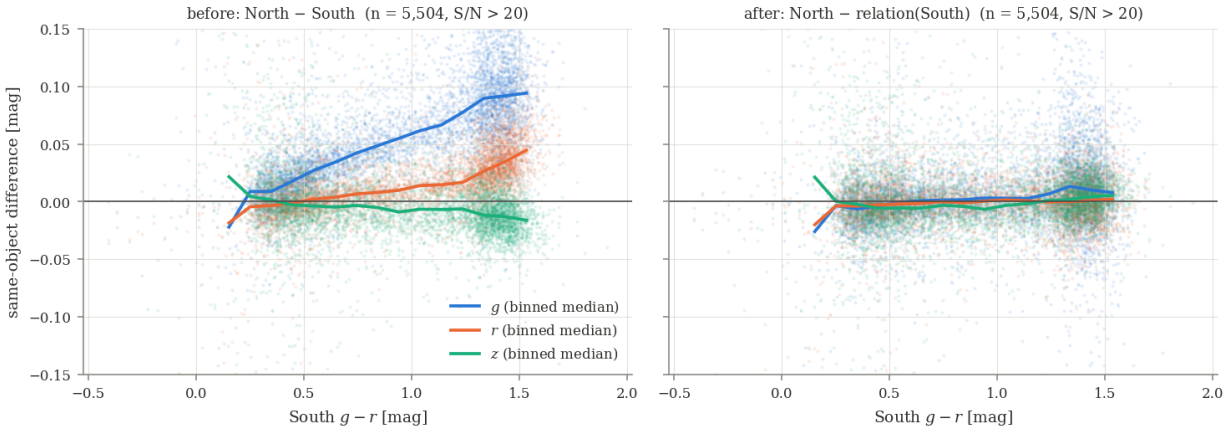


Figure 6: Same-object North minus South luptitudes in overlap fields ($S/N > 20$ in all six bands), against South $g - r$. Left: raw difference. Right: after applying the DESI relation to the southern values. Lines: binned medians.

6 The quasar model

6.1 Redshift slices and extreme deconvolution

For each of the 43 slices, the latent quasar luptitudes follow a Gaussian mixture with K components ($K = 20$ for 23 slices, 12 for 6, 8 for 13, and 4 for the sparsest one, 640 objects at $z = 4.35$):

$$p(\boldsymbol{x} \mid Q, z_k) = \sum_{i=1}^{K} \alpha_i \, N(\boldsymbol{x}; \boldsymbol{\mu}_i, V_i). \tag{7}$$

Extreme deconvolution (Bovy et al., 2011a) fits $\alpha_i, \boldsymbol{\mu}_i, V_i$ to noisy, incomplete data. Each object contributes $N(\boldsymbol{y}; H\boldsymbol{\mu}_i + \boldsymbol{b}, HV_iH^{\mathrm{T}} + T + \boldsymbol{C})$ in its observed bands only, so measurement errors are removed from the fitted widths rather than absorbed into them. Section 8 describes the fitting. At scoring time the candidate's own covariance is added back, and the density is conditioned on the reference band: $p(\boldsymbol{y}_{\mathrm{other}} \mid m_{\mathrm{ref}}) = p(\boldsymbol{y})/p(m_{\mathrm{ref}})$. The scorer interpolates linearly between neighbouring slices in redshift.

6.2 Two quasar models: with and without magnitude dependence

We deliver two complete models. They are identical in everything (bands, observation model, extinction, background, catch-all, priors, faint limit and support procedure) except one assumption about quasars: whether a quasar's colours at fixed redshift may depend on its apparent magnitude. The pointer `current` names the magnitude-dependent model, which performs better on our tests; this is a convention, not a verdict that the other model is inferior. The magnitude-independent model, `current_magindep`, should be used alongside it.

Conditioning a joint mixture on m_{ref} lets a model's colour distribution depend on apparent magnitude: a component's colour mean and width shift with m_{ref} unless its covariance forbids it. Apparent magnitude enters the likelihood in both models through the abundance prior $\Sigma_Q(z, m)$; the question is only whether it also changes the predicted colours.

Magnitude-independent model. This is the XDQSO assumption (Bovy et al., 2011b). We rewrite the latent vector as one magnitude and 35 colours, $u_r = x_r$ and $u_j = x_j - x_r$ for $j \neq r$, with r the shared Legacy r coordinate. In every component the magnitude coordinate has the same fixed distribution, $N(20, 100\,\mathrm{mag}^2)$, and zero covariance with the colours, so conditioning on the reference band leaves the colour likelihood independent of brightness (the residual dependence through the finite magnitude variance is below 10^{-3} nat for a ± 3 mag shift). With this block structure the likelihood factorises, and the exact constrained M step is the ordinary update of the colour block (Appendix B); a unit test confirms that it equals a fit to the colours alone.

Magnitude-dependent model. The same mixture without the constraint: magnitude and colours are fitted jointly, so each slice's colour distribution may shift and change width with brightness.

What each assumes, and its risk. The training quasars were selected by DESI and SDSS with colour cuts that change with magnitude, and quasar populations themselves change with luminosity. The magnitude-dependent model can describe any real dependence, but it can also learn the selection as if it were quasar physics and then penalise a genuine quasar whose colours fall outside the selection at its magnitude. The magnitude-independent model cannot learn that selection, but it must use one colour width for all magnitudes, so where the observed colour scatter changes with brightness it is too broad for bright quasars and too narrow for faint ones (Section 10.1). Our test quasars share the training selection, so the tests measure the cost of the magnitude-independent assumption on selected quasars, not its benefit for quasars outside the selection.

Measured differences (Section 10; held-out rows, faint limit applied; density of the magnitude-dependent model relative to the magnitude-independent one):

quantity	mag.-independent	mag.-dependent
held-out quasar density, per slice	—	+0.67 nat (0.38–0.88)
quasars inside the 50%/90% contours, $17 \leq r < 19$	0.62/0.94	0.51/0.90
quasars inside the 50%/90% contours, $21.5 \leq r < 22.5$	0.41/0.86	0.49/0.89
AUC, all magnitudes	0.9934	0.9939
AUC, $r < 18$	0.986	0.982
AUC, $18 \leq r < 19$	0.998	0.995
AUC, $r > 21$	0.977–0.994	0.987–0.994
AUC at $z_0 = 3.45$	0.996	0.998
quasars with $p_{\mathrm{quasar}} > 0.5$	91.2%	94.0%
unflagged background with $p_{\mathrm{quasar}} > 0.5$	0.33%	0.44%
quasars not ranked	0.57%	0.61%
grid points with $p_{\mathrm{quasar}} > 0.5$, no support cut	2	1

With the calibrated outside-both-models test (Section 9.2) the two models rank quasars against background objects equally well overall. A paired bootstrap over sky regions gives $\Delta\mathrm{AUC} = +0.0005$ (95% interval 0.0000 to +0.0015, marginal); the magnitude-independent model is slightly better at $18 \leq r < 21$ (-0.003, interval -0.005 to -0.001, at $18 \leq r < 19$; -0.001 at $19 \leq r < 21$) and the magnitude-dependent one at $r \geq 21$ (+0.003, +0.002 to +0.004). The magnitude-dependent model gives $p_{\mathrm{quasar}} > 0.5$ to +2.7 (2.3–3.5) percentage points more test quasars, and to +0.11 (0.01–0.21) percentage points more unflagged background objects. Under the earlier fixed 4σ test the magnitude-dependent model appeared much better for bright quasars ($\Delta\mathrm{AUC} = +0.020$ at $r < 18$); most of that difference came from the magnitude-independent model refusing more bright quasars under that test, not from its ranking.

Which to use. For candidates that resemble the training quasars, the magnitude-dependent model recovers more quasars at $p_{\mathrm{quasar}} > 0.5$, and the two rank equally well. For candidates that may lie outside the DESI and SDSS selection, for example quasars with unusual colours for their brightness, the magnitude-independent model makes the weaker assumption, and the tests above cannot reveal that advantage because the test quasars share the training selection. Use both: score every candidate with `current` and `current_magindep`. A candidate ranked highly by both is robust to this assumption; a large disagreement marks a case that depends on it and should be reported as such.

7 The background model

Design. Faint sources dominate the background counts (69% of fit/select rows have $r > 23.5$). A single mixture over all magnitudes therefore spends almost all its components on faint stars and describes bright ones poorly. Following XDQSO (Bovy et al., 2011b), which fits stars in narrow magnitude bins, we fit the background separately in bins of Legacy r. On held-out stars this describes the data better than one 20-component mixture over all magnitudes in every bin, by 4.9 nat per object at $r < 17$ and 0.2 at $r > 23$. Sharing one set of colour shapes across magnitude, with only their weights free, is worse at $r > 22$: the shape of the stellar locus itself changes with magnitude. CondXD (Kang et al., 2025) reaches comparable results with a neural network that makes all mixture parameters functions of magnitude.

Binned fit. The fit/select background rows above the faint limit are split into eight bins of Legacy r: 15.5–17, 17–18, …, 22–23 and 23–24.5. Each bin is fitted independently by extreme deconvolution with $K = 20$ components over the 36 latent coordinates (written as r and 35 colours), on its own rows plus those within 0.25 mag of its edges (19,158 to 255,709 rows per bin), with the

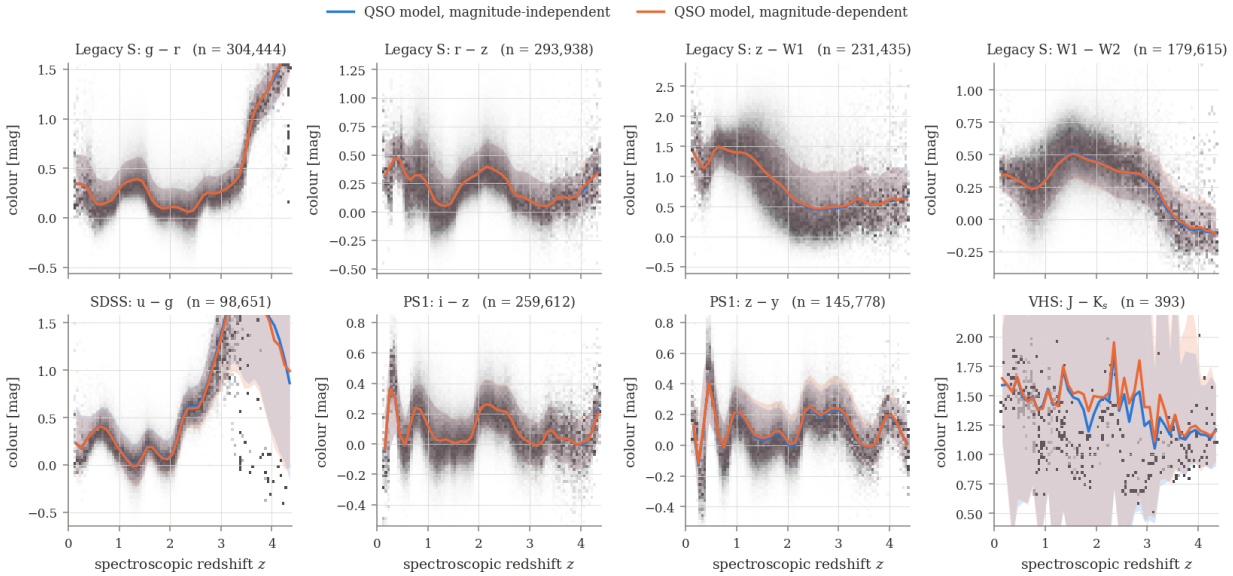


Figure 7: Quasar colour against redshift. Grey: test quasars with both bands at $S/N > 10$ (density normalised in each redshift column of width 0.05). Lines and bands: per-slice median and 16–84% range of the noise-free model colour, magnitude-independent (blue) and magnitude-dependent (orange, marginalised over magnitude). The data include measurement noise and a signal-to-noise selection that the model bands do not.

MAP update and predictive early stopping of Section 8 on 1,000 test-role objects per bin, starting from equal weights. The eight mixtures are joined into one 160-component mixture, each bin's weights scaled by its share of training objects. Conditioning this mixture on $m_{\rm ref}$ weights each component by $N(m_{\rm ref}; \mu_{r,i}, V_{rr,i})$, which selects the bin and blends neighbouring bins smoothly at the edges. On 17,937 held-out stars this beats picking the single bin that contains $m_{\rm ref}$ by 0.12 ± 0.015 nat per object overall (0.24 ± 0.07 within 0.1 mag of a bin edge, 0.08 ± 0.01 further than 0.3 mag), in both hemispheres and in every bin (from $+0.002 \pm 0.005$ at $22 \le r < 23$ to $+0.23$). Because $m_{\rm ref}$ is Legacy r for every scored source, routing and conditioning use the same magnitude.

Training sample cleaned of quasars. Known quasars (DESI DR1, DR16Q) were removed from the background sample when it was built, but it still contains unidentified ones, increasingly at faint magnitudes: among background rows detected in $W1$ and $W2$, the fraction with the AGN colour $W1 - W2 > 0.8$ (Vega; Stern et al. 2012) is 3% at $19 \le r < 20$ and 37% at $22 \le r < 23$. A binned model is flexible enough to learn them as background, which lowers the recall of real quasars (in a test, from 0.83 to 0.74 at $p_{\rm quasar} > 0.5$). We therefore remove from training (and from the background test sets) members of Quaia (Storey-Fisher et al., 2024) matched by Gaia DR3 `source_id`, and sources with $W1 - W2 > 0.8$ (Vega) at $S/N \ge 5$ in both bands, except very red optical sources ($r - z > 2$, kept as brown-dwarf candidates). This removes 0.1% of training rows at $r < 18$ and 3–5% at $r > 20$. A Gaia zero-proper-motion cut was tested and rejected: 33–56% of zero-motion sources have stellar $W1 - W2$ even at $\sigma_\mu < 0.2$–$0.5\,{\rm mas\,yr^{-1}}$ (distant halo stars), and Gaia does not reach the faint bins where contamination matters. Faint unidentified quasars not detected by WISE remain; we accept this.

Sky dependence. Component shapes are the same everywhere; the weights vary with position as a softmax gate,

$$w_i(\ell, b) \propto \exp[a_i + \boldsymbol{\beta}_i \cdot \boldsymbol{f}(\ell, b)]\,, \qquad \boldsymbol{f} = (\csc|b|,\ \cos\ell,\ \sin\ell,\ \csc|b|\cos\ell,\ \mathrm{sign}\, b)\,, \tag{8}$$

fitted by maximum likelihood with fixed component densities and an L2 penalty of 1 on $\boldsymbol{\beta}$, on 300,000 cleaned training rows. The gate is fitted to the joint density, but it is judged by the

quantity the scorer uses: on 60,000 held-out rows in the test cones, with the final background, it improves the conditional density $\ln p(\boldsymbol{y}_{\rm other} \mid r)$ over one all-sky weight vector by 0.0260 ± 0.0015 nat per object, positively in every magnitude and $|b|$ bin, and beats a HEALPix hierarchy of cell weights (nside 4 pooled to nside 2 and all-sky, $n_0 = 1000$: 0.0219 ± 0.0011). The scorer evaluates the gate at nside-16 pixel centres (3.7°; no measurable loss). The effect is small because stellar colours at $|b| \geq 25°$ change little with position.

Surface density. $\Sigma_B(m)$ is measured per reference band from counts in regions of measured usable area, pooled with $n_0 = 100$, on the cleaned sample, so that counts and colour model describe the same population (removing the flagged sources lowers it by less than 0.5% at $r < 18.5$ and by 3–6% at $20 \leq r \leq 24.5$). A candidate-local refit of weights and density in a declared cone is available.

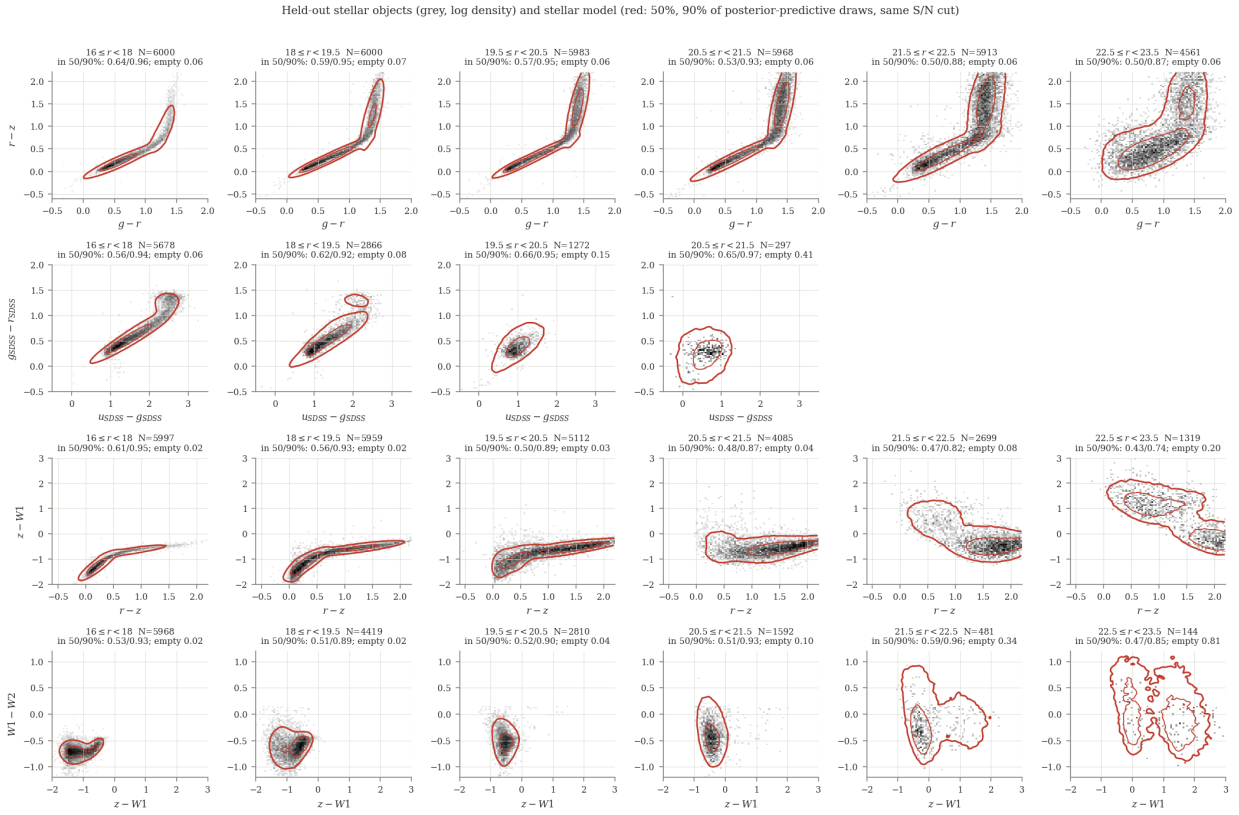


Figure 8: Background colour–colour density in six bins of Legacy r and four planes. Grey: test-cone background objects above the faint limit, likely quasars removed, all plane bands at $S/N \geq 5$. Red: 50% and 90% contours of posterior-predictive draws of the binned background for the same objects (200 per object, conditioned on each object's r and sky position, with the object's own flux errors added in flux space and the same $S/N \geq 5$ selection). Titles: data inside the 50%/90% regions (ideal 0.50/0.90) and the fraction of draws in empty cells. Panels with fewer than 100 objects are left empty.

8 Fitting

8.1 The covariance update

The extreme-deconvolution E step computes each component's responsibility q_{ni} for object n, and the posterior moments of $\boldsymbol{x}$. The M step updates the weights, the means, and the covariance from the scatter S_i about the new mean. We use the conjugate-prior MAP update of Bovy et al. (2011a)

(their eqs. 19–20):

$$V_i = \frac{q_i S_i + \nu w I}{q_i + \nu}, \qquad q_i = \sum_n q_{ni}, \quad \nu = 1,\ w = 10^{-3}\,\mathrm{mag}^2, \tag{9}$$

which maximises the log posterior with prior $-\frac{\nu}{2}\left[\ln|V_i| + w\,\mathrm{tr}(V_i^{-1})\right]$ per component, so EM cannot decrease the reported objective. (A fixed additive floor, $V_i = S_i + wI$, maximises no objective, and in practice lets the likelihood decline.)

8.2 Predictive early stopping

Every 20 updates we compute the held-out density on a fixed stopping panel: up to 512 test-role objects per hemisphere per fit, recorded and excluded from later validation. A fit stops when a block gains less than 0.02 nat per object, when the objective changes by less than 10^{-4} (relative), or after 300 updates. The best checkpoint, including the warm start, is kept. The 0.02 nat threshold is about twice the sky-bootstrap standard error of late block gains. This is predictive early stopping, not convergence. Sparse slices benefit most: where held-out density falls in the first block, as at $z = 4.05$, the fit keeps its warm start. All 43 quasar fits of each model stopped this way, with no objective decrease (Fig. 9a,b). The eight background bins use the same rule on 1,000 test-role objects per bin; they kept checkpoints at 40–92 updates (Fig. 9c).



Figure 9: (a) Update at which each quasar-slice fit kept its checkpoint, against slice redshift. (b) Held-out gain of that checkpoint over the fit's warm start. (c) Held-out density gain over the warm start against update for the eight background bins (Legacy r); stars mark the kept checkpoints.

9 Priors, catch-all, support and the outside-both-models test

Abundance priors. $\Sigma_Q(z, m)$ and $\Sigma_B(m, \ell, b)$ are stored per reference band, per unit native luptitude; Legacy r is the reference band of every scored object, so only its two priors (South and North) enter. Background densities are counted from corrected photometry (Section 7). The quasar abundance is described in Section 9.1.

9.1 Quasar abundance prior

$\Sigma_Q(z, m)$ is the number of quasars per deg^2, per unit redshift and per magnitude. It multiplies every quasar likelihood (Eq. 3), so it sets p_{quasar}, $\log \mathcal{R}$ and the blend-mode magnitude priors. Since 5 October 2026 it is the eBOSS quasar luminosity function, not spectroscopic counts.

Earlier construction and its problem. Until 5 October 2026 the prior was built from spectroscopic quasars: a flat-in-redshift draw of the de-duplicated DESI DR1 and DR16Q parent at $|b| \geq 25°$, weighted back to the parent's redshift distribution, counted in each band's footprint and divided by its area, then multiplied by one completeness constant $C = 2.37$. C was chosen

so that the Legacy South r prior integrated, over $17 < r < 22.5$, to an earlier DESI-only prior normalised to DESI's density in fully covered sky ($240\,\mathrm{deg}^{-2}$); the point-source recount kept the same C, and the corrected-magnitude version shifted it by the mean extinction (0.094 mag in r). Figure 10 compares this prior with the completeness-corrected eBOSS quasar luminosity function (Palanque-Delabrouille et al., 2016). The spectroscopic counts themselves (the prior divided by C) are below the luminosity function at every magnitude, by a factor that depends strongly on magnitude: 0.8–1 at $r < 18.5$, where SDSS is nearly complete, 0.6 at $19 < r < 21$, 0.26 at $r = 22.25$ and 0.03 at $r = 23.25$, where spectroscopic targeting ends. One constant cannot correct that: the prior was 1.4–3 times too high at $r < 20$ (4–9 times for $z > 2$), within ±50% at $19.5 < r < 22$, and 0.6, 0.3, 0.08 and 0.02 of the luminosity function at $r = 22.25$, 22.75, 23.25 and 23.75. Under the faint limit we score sources to $r \approx 23.5$, so faint quasars were given probabilities far too low.

Construction. The luminosity function of Palanque-Delabrouille et al. (2016) is measured from 13,876 quasars selected by variability (not colour) over $94.5\,\mathrm{deg}^2$ of Stripe 82, with completeness corrections for selection, fibre assignment and redshift failures derived from an independent colour-selected control sample. We use its best-fit PLE+LEDE model through the paper's Table 7: predicted quasar counts in (extinction-corrected) r per $10{,}000\,\mathrm{deg}^2$, in bins of 0.5 mag and $\Delta z = 1$. For each magnitude row, the cumulative counts at $z = 0, 1, \ldots, 5$ are interpolated with a monotone cubic (PCHIP), and each cell of our redshift grid ($\Delta z = 0.1$, $0.1 < z < 4.4$) receives the difference across its edges, so every published bin total is preserved and the density is smooth and positive. Between rows the log density is interpolated linearly in magnitude, and the last slope is continued beyond $r = 23.75$. SDSS r and Legacy r differ by a few hundredths of a magnitude for quasars; this is ignored. Colour models, background densities and all other inputs are unchanged (`scripts/method_unified/build_qlf_prior.py`).

Checks. The prior reproduces the published counts within 3% per magnitude at $17 < r < 23.75$, and within 0–10% per redshift bin; it is lower only for $z > 4.4$, outside our grid. Inside each $\Delta z = 1$ bin the interpolated redshift shape follows the shape of the spectroscopic counts where those are reasonably complete ($19.5 < r < 21.5$) to 10–25%; the largest differences, 40–65%, are near $z \approx 3.3$–3.6, where the spectroscopic counts dip because quasar colours cross the stellar locus and colour selection loses them. For $0.15 < z < 1$, where the luminosity function is extrapolated below its fitted $z = 0.68$ and many quasars look extended, it agrees with the spectroscopic point-source counts to 10% at $18 < r < 21$. On the faint end, the background test sample (catalogue objects minus known quasars) must contain the quasars without spectra: at $22 < r < 24$ the luminosity function minus the known quasars implies that 1.7–3.0% of it are quasars. With this prior, 1.3–1.8% of those background objects receive $p_{\mathrm{quasar}} > 0.5$; with the earlier prior 0.0–1.4%. The extra faint objects called quasars are therefore fewer than the quasars known to be hiding in the background sample (Section 10.3).

Caveats. The luminosity function is fitted for $0.68 < z < 4$ and $g < 22.5$ ($r \approx 22.3$); fainter and at lower redshift it is the published model extrapolated, and its two fits (PLE and PLE+LEDE) differ by up to 16% at $r = 23.75$. The shape within each $\Delta z = 1$ bin is an interpolation, not a measurement. It counts all quasars, while we score point sources; the agreement at $0.15 < z < 1$ suggests this matters little above our faint limit. The background density Σ_B is counted from the catalogue with known quasars removed, so at $r > 22.5$ it still holds the 2–3% unlabelled quasars; their intensity is counted twice, once in Σ_Q and once in Σ_B, which lowers faint p_{quasar} by up to about 3%.

Catch-all. U is a Student-t with $\nu = 4$ degrees of freedom, centred on the background mean, with the binned background mixture's second-moment matrix as its scale. Measurement noise is convolved exactly through the t's Gaussian scale-mixture representation. Its share η is a Beta-MAP

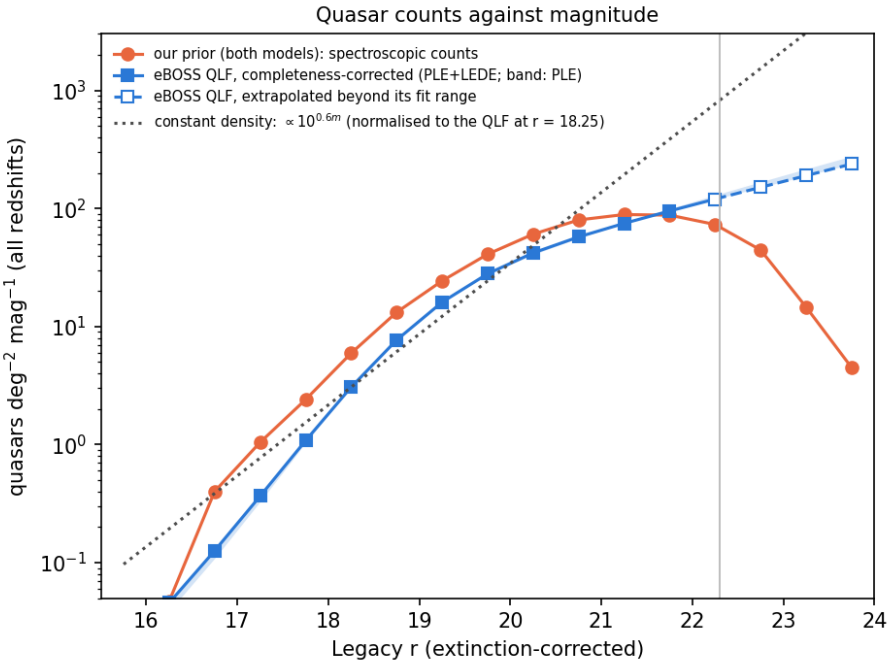


Figure 10: Quasar counts per deg^2 per magnitude (all redshifts) against extinction-corrected Legacy r: the earlier prior (spectroscopic counts times one constant), the eBOSS luminosity function now used (solid where fitted, dashed where extrapolated; band: PLE against PLE+LEDE), and constant comoving density in Euclidean space ($\propto 10^{0.6m}$, normalised at $r = 18.25$).

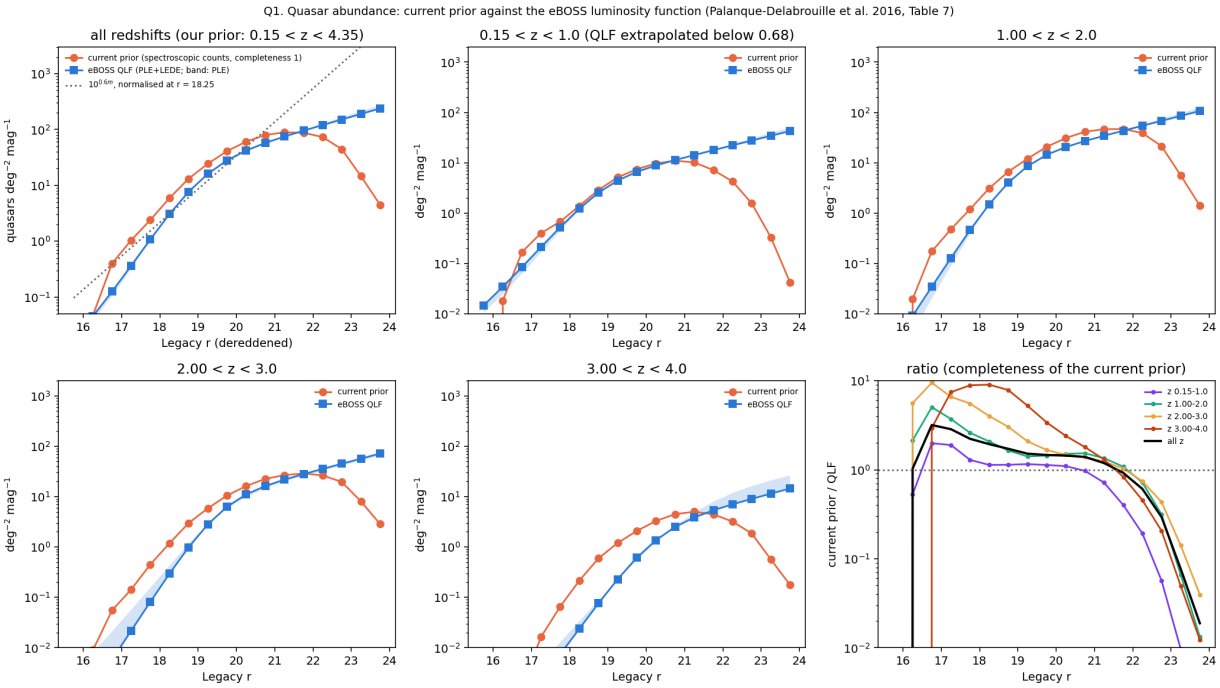


Figure 11: The earlier prior against the eBOSS luminosity function in redshift bins, and their ratio (bottom right), which is the completeness the single constant was meant to correct.

fit per reference band and three magnitude bins, on 148,853 calibration background objects above the faint limit with likely quasars removed. The global share is 0.14%.

Support cut. For a candidate, z_0 and the observed bands, the scorer draws 255 simulated quasars from the conditional model at z_0, adds the candidate's errors, and ranks the candidate's conditional density among them. Candidates below a threshold are not ranked; their likelihoods are kept as diagnostics. The percentile depends on the reference band, so the threshold was recalibrated under the faint limit on 2,000 calibration quasars (all bands and Legacy g, r, z only; 3,890 evaluations). At the 99.5% target it is $2/256 = 0.0078$, retaining 99.64% and 99.59% of calibration evaluations. Fig. 12 shows the trade-off on held-out rows: the cut at 2/256 removes the 2 (magnitude-independent) and 1 (magnitude-dependent) high-quasar points that the artificial grids place outside the models' dense core, at a cost of 0.5% of test quasars and of 0.002 in the AUC of p_{quasar} (0.992 to 0.990); stricter cuts lose quasars faster. (Under the earlier fixed 4σ outside-both test the same cut removed 12 and 4 stray points and raised the AUC, because that test let more extrapolated objects through at few bands.)

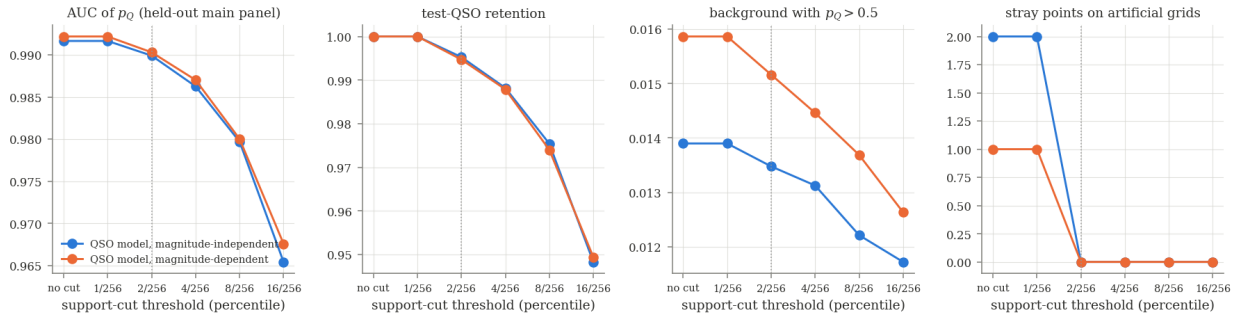


Figure 12: Support-cut threshold (percentile; dotted: the calibrated 2/256) on the held-out main panels of Section 10.3 and the artificial grids of Section 10.4: AUC of p_{quasar} (the scorer does not compute $\log \mathcal{R}$ for rejected objects, so p_{quasar} ranks all thresholds alike; unranked objects at the bottom), fraction of test quasars kept, fraction of background objects with $p_{\mathrm{quasar}} > 0.5$, and grid points with $p_{\mathrm{quasar}} > 0.5$ in the low-density mask.

9.2 Outside both models: a calibrated test

Statistic. For population c (quasars at any redshift, or the background) the scorer computes the nearest-component distance of the object's non-reference bands given its reference band,

$$T_c = \min_j \left(\boldsymbol{y}_O - \boldsymbol{\mu}_{cj,O|r}\right)^{\mathrm{T}} \mathbf{C}^{-1}_{cj,O|r} \left(\boldsymbol{y}_O - \boldsymbol{\mu}_{cj,O|r}\right),$$

where O are the observed non-reference bands and $\boldsymbol{\mu}_{cj,O|r}$, $\mathbf{C}_{cj,O|r}$ are the mean and covariance of component j conditioned on the object's reference luptitude, after the object's measurement covariance has been added. For quasars j runs over all components of all 43 slices; for the background over the 160 components with their weights at the object's sky position. $\sqrt{T_c}$ is reported as `qso_ood_sigma_any_z` and `bkg_ood_sigma`.

Why a fixed cut fails. Until 5 October 2026 an object was outside both models when $\sqrt{T_Q} > 4$ and $\sqrt{T_B} > 4$. For one Gaussian in k dimensions $T \sim \chi^2_k$, so a member typically lies at $\sqrt{k}$: 3.2σ at $k = 10$ and 4.5σ at $k = 20$. A fixed cut is strict for objects observed in many bands and lenient for objects observed in few. On the held-out test panel it refused 2.5–2.9% of real quasars, rising from 0% at fewer than 12 observed bands to 7–15% at 20 or more, and background objects in the same way (Fig. 13). The refused quasars continue the distance distribution of the ranked ones just past the cut; rescored with Legacy $g, r, z, W1, W2$ alone, 93–94% of them are ranked. The textbook χ^2_k correction does not fit either: the minimum over many overlapping components is much smaller

than χ^2_k, and converting it with k degrees of freedom would leave almost nothing flagged. The full diagnosis is in `docs/pquasar_diagnostics`.

Calibrated test. The distribution of T_c is computed for each object by simulation, in its own observed bands, with its own noise and its own reference value. The scorer draws $B = 256$ members of population c from the model conditioned on the object's reference luptitude (components reweighted by their likelihood at that value; quasar slices with equal weight), each including the object's measurement noise, computes the same statistic for each, and returns

$$p_c = \frac{1 + \#\{b : \; T_c^{(b)} \geq T_c^{\mathrm{obs}}\}}{B + 1},$$

the probability that a member of population c lies at least as far from every component. An object is outside both models when $p_Q < \alpha$ and $p_B < \alpha$, with $\alpha = 2/256$, the same level as the support cut, fixed before any test. This is a parametric-bootstrap tail probability of the existing statistic: it accounts for the number and the identity of the observed bands, the noise, and the minimum over components, with no fitted correction. Draws are seeded from the object's own inputs, so results do not depend on batch order. It costs about 13 ms per object for both populations. The scores record p_Q and p_B as `qso_ood_p` and `bkg_ood_p`; `UnifiedPSFModel.score(..., ood_calibration=None)` restores the fixed cut.

Checks. For members of a known Gaussian mixture, p is uniform in 4, 12 and 25 dimensions, where the raw distance of a typical member already exceeds 3σ (unit tests). On 4,000 calibration-role quasars and 4,000 background objects the test flags 0.23–0.25% and 0.48–0.60%; the fraction of each population with its own $p_c < \alpha$ is 0.4% (quasars) and 0.75% (background), against the nominal 0.78%. On real test quasars the p_Q values lie below the uniform line (Fig. 13, right): the quasar model is slightly broad, and the test is conservative. On the test panel the refusal rate no longer depends on the number of bands. The performance numbers of Section 10.3 and the calibration of Section 10.6 use the calibrated test.

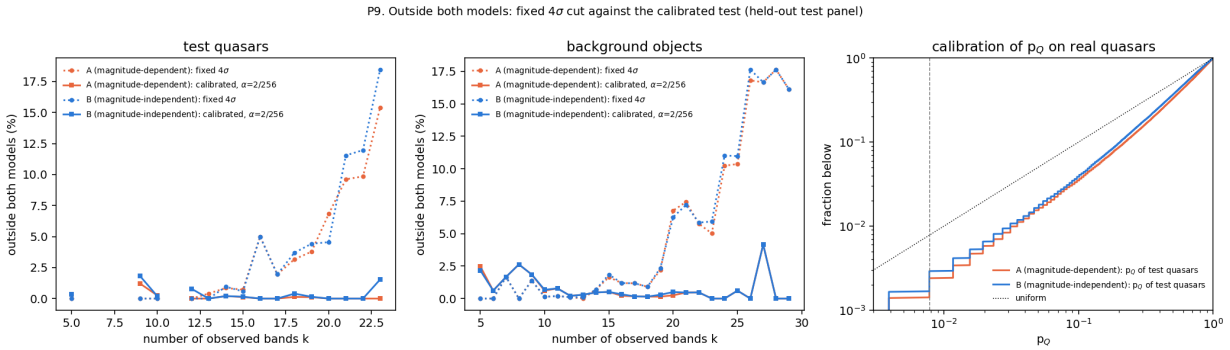


Figure 13: Outside both models, against the number of observed bands k, for the earlier fixed 4σ cut (dotted) and the calibrated test at $\alpha = 2/256$ (solid), for held-out test quasars (left) and background objects (middle), both models. Right: cumulative distribution of p_Q for the test quasars (dotted: uniform; dashed: α).

10 Validation

Tests use test-role objects above the faint limit and outside every recorded development exclusion.

10.1 Colour–colour distributions

Figures 14–16 compare model densities with the data in two colour planes: Legacy South $(g-r, r-z)$ and $(r-z, z-W1)$. Model contours are conditioned on each bin's median r, weighted by the

data's redshift mix, and convolved with the bin's median measurement covariance, so they can be compared directly with the data. Each panel title gives the fraction of the data inside the model's 50% and 90% regions; a well-calibrated model gives 0.50 and 0.90. The magnitude-dependent quasar model gives 0.48–0.51 and 0.89–0.90 in every magnitude bin and both planes. The magnitude-independent model must use one width for all magnitudes. It is too broad for bright quasars (0.62/0.94 at $17 \leq r < 19$) and too narrow for faint ones (0.41/0.86 at $21.5 \leq r < 22.5$). The apparent colour scatter of quasars, after deconvolution, grows towards faint magnitudes in this sample. Underestimated photometric errors, a change in the population, and selection can all produce this. The binned background follows the stellar locus at every magnitude and sky position: in $(g-r, r-z)$ it gives 0.43–0.51 and 0.89–0.92 in the magnitude bins and 0.42–0.46 and 0.89–0.90 in the sky bins. In $(r-z, z-W1)$ the 90% coverage is 0.82–0.88; part of this is the $W1$ detection selection of the plotted data, which Fig. 8 reproduces in the model and which brings the coverage to 0.87–0.95 for $r < 21.5$.

10.2 Held-out density

Per redshift slice, the magnitude-dependent quasar model's held-out density is higher than the magnitude-independent model's by a median 0.666 nat per object (range 0.381–0.876): the cost of the magnitude constraint in density (Section 6.2). The background's fit is shown directly in Fig. 8.

10.3 Performance across sky, magnitude, redshift and band coverage

These tests use 20,000 quasars and 20,000 background objects per hemisphere, drawn before the faint limit from test-role rows outside every development exclusion and outside the rows used to choose support thresholds; 38,881 quasars (97.2%) and 14,247 background objects (35.6%) pass the faint limit. Background objects are scored at redshifts drawn from the test quasars' redshifts. The four metrics are those of Section 2: AUC, held-out density, support retention, and high-QSO incidence, the fraction of background objects ranked with $p_{\mathrm{quasar}} > 0.5$. The background test objects include unidentified quasars, so incidence is not a contamination rate.

Overall. The AUC is 0.9934 (magnitude-independent) and 0.9939 (magnitude-dependent); South 0.992 and 0.993, North 0.995 for both. The support cut keeps 99.53% and 99.47% of test quasars, in agreement with the 99.5% calibration target on rows that did not choose it. In total 0.57% and 0.61% of test quasars are not ranked: 0.38% and 0.45% at the support cut, 0.17% and 0.14% outside both models (Section 9.2), and 0.02% with no quasar prior at their magnitude. Of the test quasars, 91.2% and 94.0% receive $p_{\mathrm{quasar}} > 0.5$. High-QSO incidence is 1.35% and 1.52% (South 0.76% and 0.90%, North 2.04% and 2.24%). Among background objects that are not flagged as likely quasars it is 0.33% and 0.44%, and the AUC against them is 0.995 for both. Compared with the earlier abundance prior (spectroscopic counts), the luminosity-function prior leaves everything at $r < 21.5$ unchanged and raises the recall at $22.5 \leq r < 23$ from 0.42 to 0.65 (magnitude-independent) and from 0.58 to 0.77 (magnitude-dependent), and at $23 \leq r < 24$ from 0.10 to 0.30 and from 0.25 to 0.54. The faint background objects it calls quasars (1.3–1.8% at $22 < r < 24$) are fewer than the unlabelled quasars the luminosity function implies for that sample (1.7–3.0%; Section 9.1); the AUC changes by -0.0004. With the earlier fixed 4σ outside-both test the AUC was 0.976 and 0.980 and 3.2% and 2.9% of test quasars were not ranked: an unranked quasar counts as ranked below every ranked background object, so recovering 2.3–2.6% of quasars raises the AUC by 0.014–0.018.

Sky position and magnitude (Fig. 17). Per-cell AUC is uniform across the footprint to within a few thousandths. Against $|b|$, AUC is 0.990–0.995 (magnitude-independent) and 0.991–0.995 (magnitude-dependent). Against reference magnitude it is 0.995–0.998 at $18 \leq r < 21$ and falls towards the faint end, to 0.977 (magnitude-independent) and 0.987 (magnitude-dependent) at $22.5 \leq r < 23.5$; at $r < 18$ (274 quasars) it is 0.986 and 0.982.

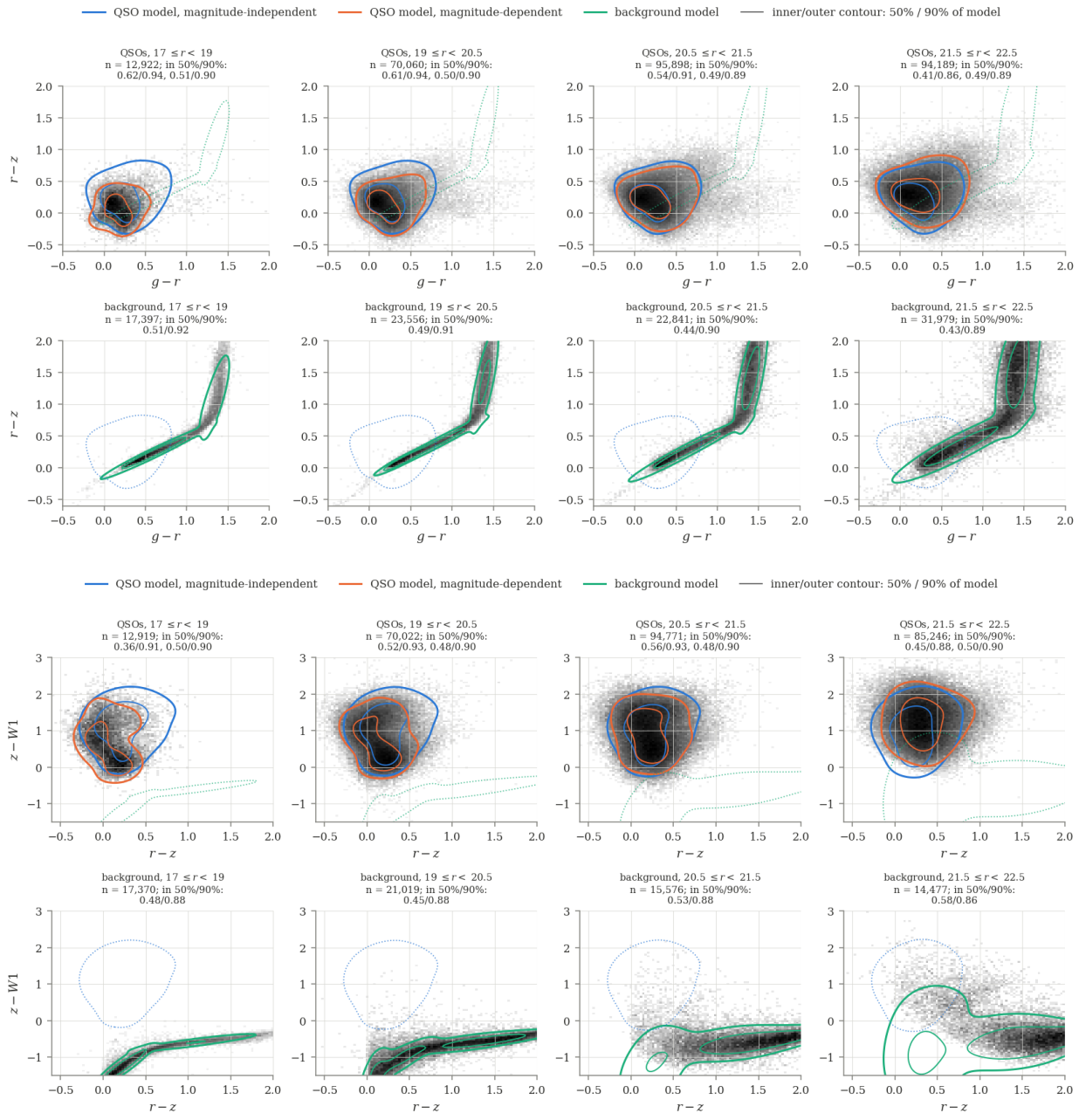


Figure 14: Colour–colour planes in bins of Legacy South r, objects above the faint limit. Top rows of each block: test quasars (grey, log density) with both quasar models (blue, orange; thick and thin contours enclose 90% and 50%) and the background model's 90% contour (dotted aqua). Bottom rows: test background objects (likely quasars removed) with the binned background and the magnitude-independent quasar model's 90% contour (dotted blue).

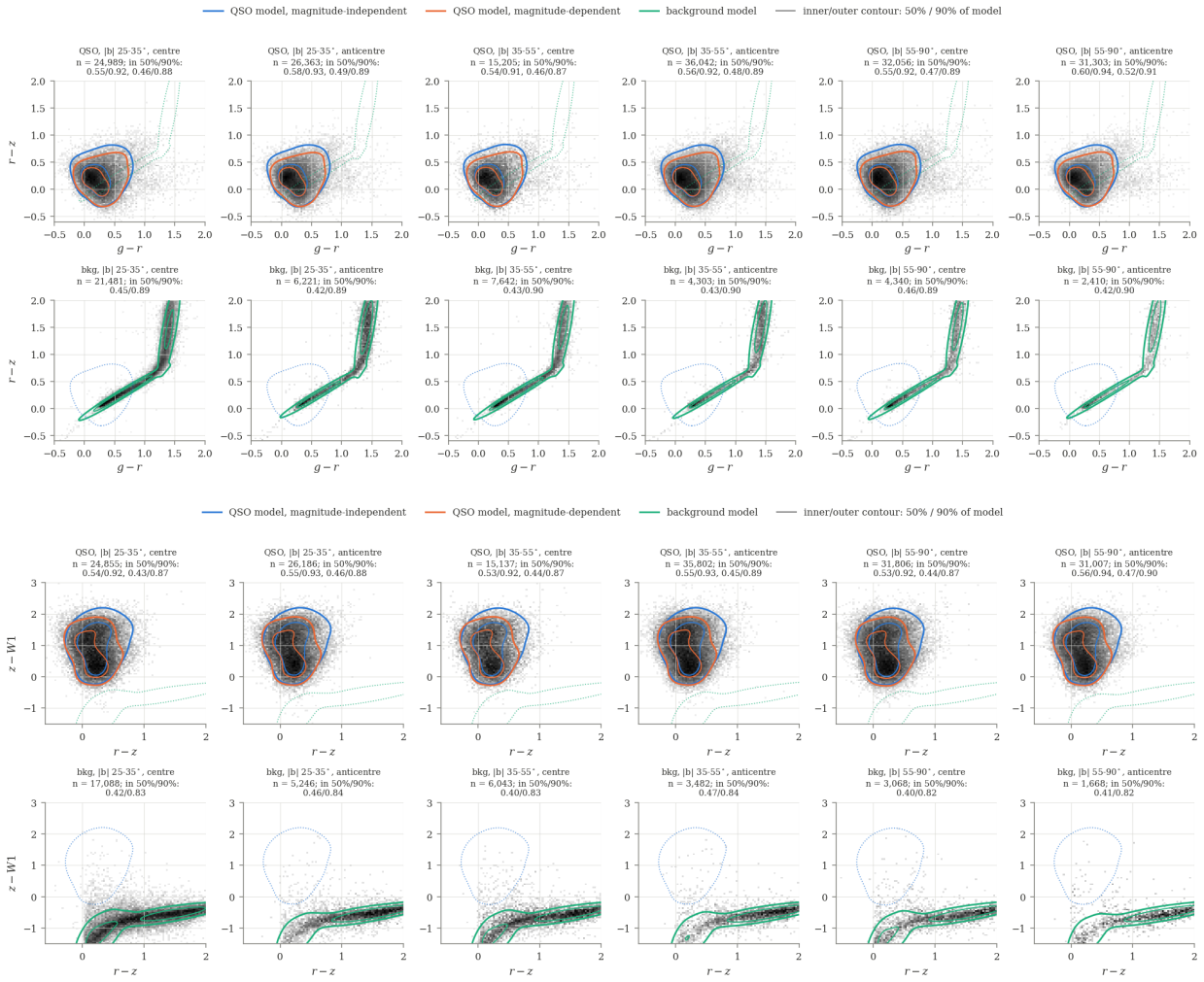


Figure 15: As Fig. 14, in bins of Galactic latitude and of longitude towards ($|\ell| < 90°$) or away from the Galactic centre, at $19 \leq r < 21.5$. Background contours use the spatial component weights averaged over each bin's objects.

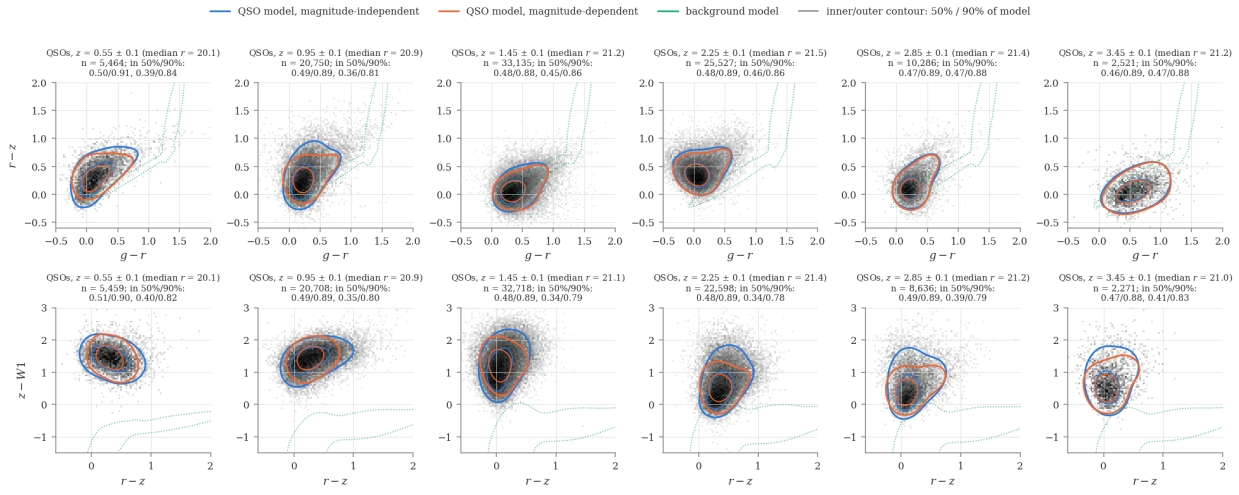


Figure 16: Quasar colour–colour planes in six redshift slices, each conditioned on the slice's median r, with the background model's 90% contour for reference.

Redshift (Fig. 18). AUC is 0.992–0.994 for both models for $0.5 \leq z_0 \leq 3$ and 0.996–0.998 at $z_0 = 3.45$. (Under the earlier fixed test it fell to 0.948 and 0.956 at $z_0 = 3.45$, where 8–15% of quasars were refused.) Held-out density falls with redshift as quasars become fainter, and the magnitude-dependent model stays higher at every redshift. Support retention is above 99% everywhere.

Band coverage (Fig. 19). Grouping objects by the surveys that observed them mixes coverage with brightness and redshift, and several groups have fewer than 150 background objects. The artificial masks score the same objects with fewer bands: Legacy g, r, z only gives AUC 0.974 and 0.975, all Legacy bands 0.993 and 0.994, all optical bands 0.989 for both. SDSS-only and PS1-only photometry has no Legacy r and was not scored here; Section 12 tests sources without Legacy r. The weakest coverage group with at least 100 quasars and 30 background objects is Legacy North+AllWISE+PS1 for the magnitude-independent model (AUC 0.985, 1,641 quasars) and Legacy South+AllWISE+PS1 for the magnitude-dependent one (0.982, 2,451 quasars).

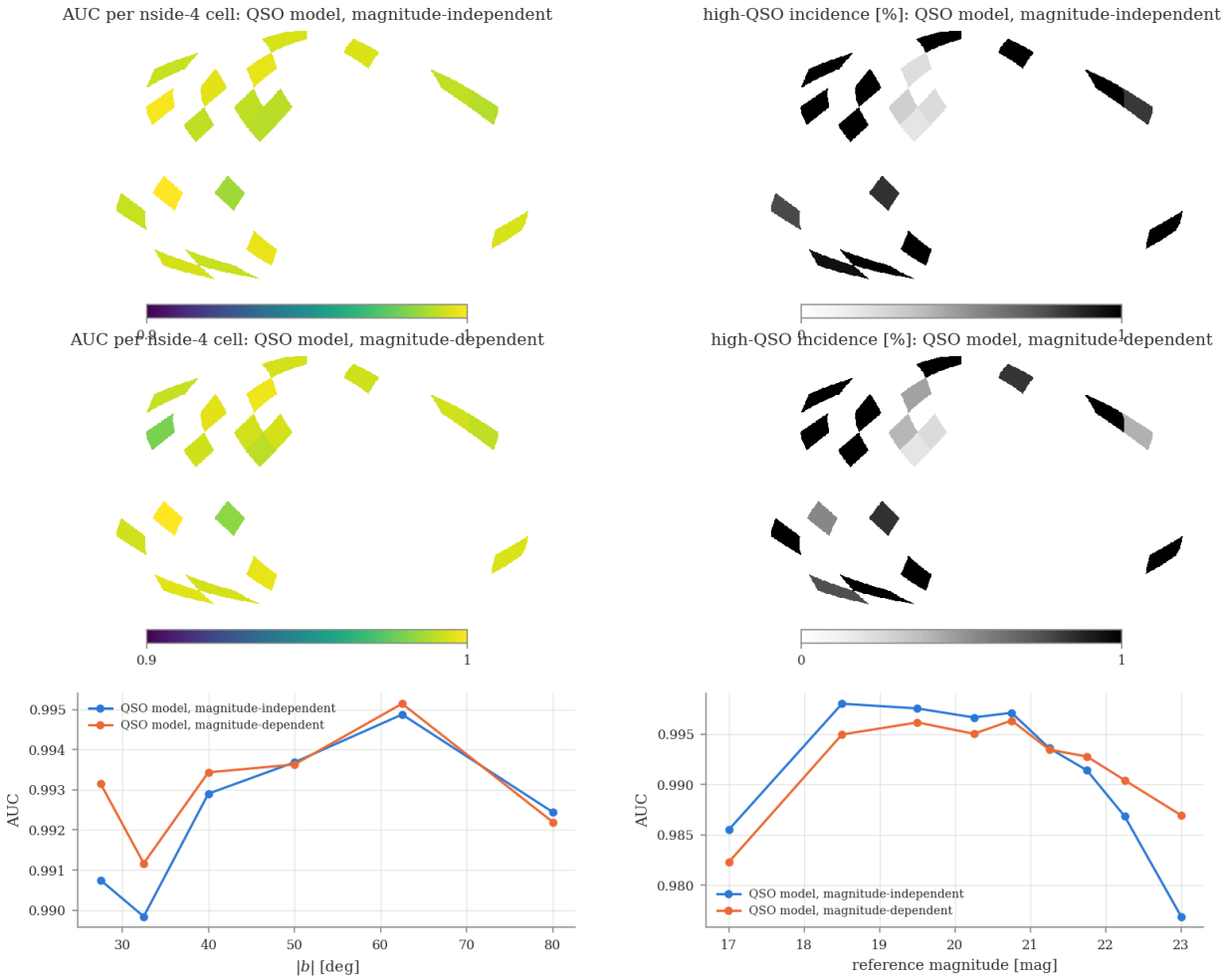


Figure 17: Performance against sky position and magnitude. Top: AUC and high-QSO incidence per HEALPix nside-4 cell (cells with at least 30 objects of each class). Bottom: AUC against $|b|$ and against reference-band magnitude.

10.4 Artificial colour grids

A colour grid places synthetic sources on a 15×15 lattice in $(g-r, r-z)$ from -6 to 8 mag at fixed r (18.5 and 21; South and North; $(\ell, b) = (180°, 45°)$; 0.03 mag errors). These are stress tests of extrapolation into empty colour space, not contamination rates. Outside the models' dense core,

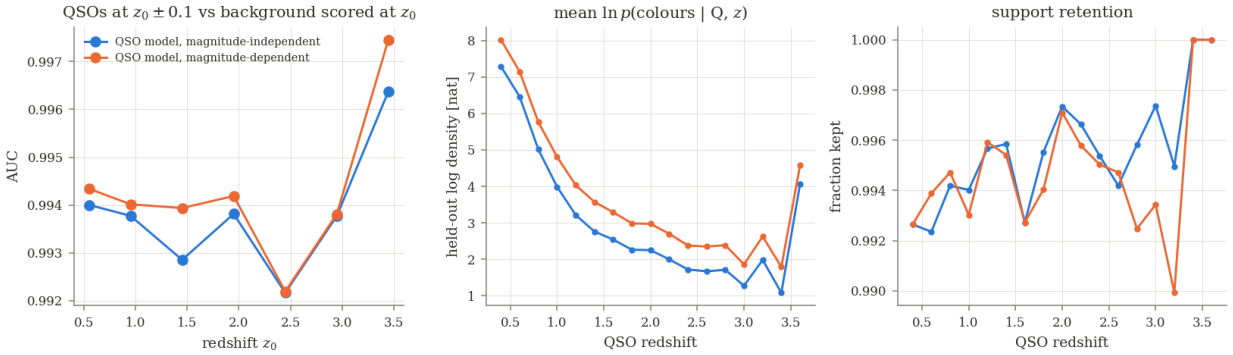


Figure 18: Performance against redshift. Left: AUC of quasars at $z_0 \pm 0.1$ against background objects scored at z_0. Middle: mean held-out log density of quasars, in bins of their redshift. Right: support retention.

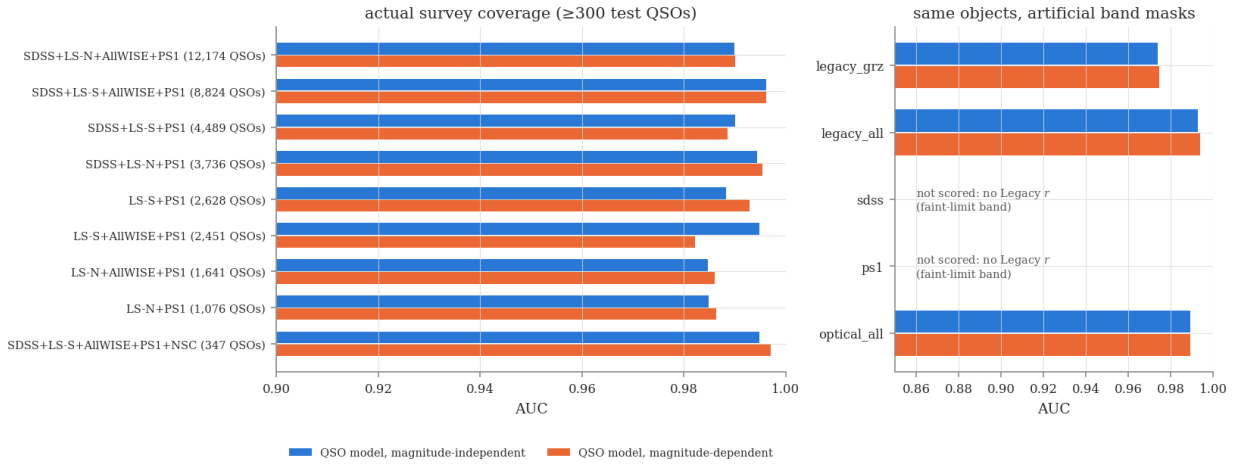


Figure 19: AUC by band coverage. Left: objects grouped by the surveys that observed them (groups with at least 300 test quasars). Right: the same objects scored with artificial band masks; SDSS-only and PS1-only objects have no Legacy r and were not scored in this test (see Section 12).

where both the quasar and the background intensity are below 1% of their grid maximum, no grid point receives $p_{\mathrm{quasar}} > 0.5$ with the support cut (Fig. 20); without it, 2 (magnitude-independent) and 1 (magnitude-dependent) would. Most of the grid is not ranked: points outside both models, or rejected by the support cut, are set aside.

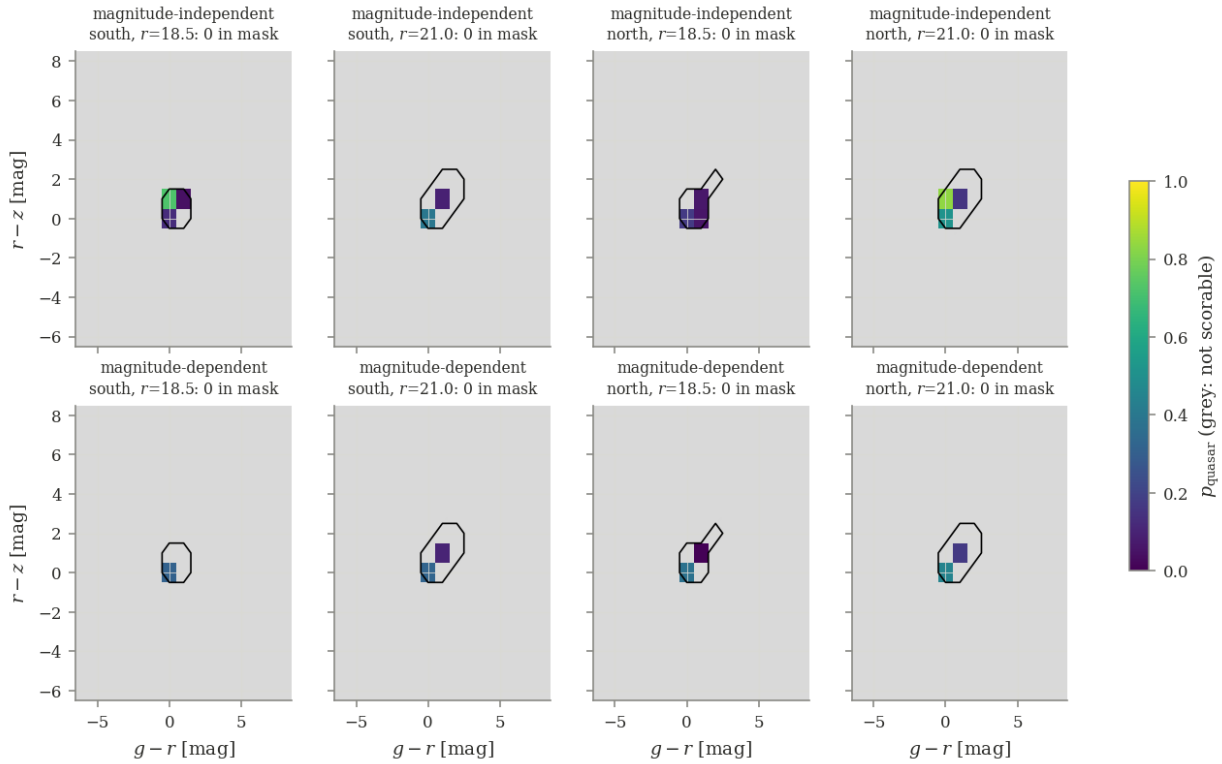


Figure 20: Total quasar probability on artificial Legacy colour grids, South and North, $r = 18.5$ and 21. Grey cells are not ranked. Black outline: the models' dense core; outside it both the quasar and the background intensity are below 1% of their grid maximum.

10.5 Worked examples

Figure 22 shows six test objects scored by both models. For each band the panels compare the observed corrected luptitude, minus Legacy r, with the model's conditional prediction at the object's redshift. A bright $z = 1.48$ quasar and a $z = 0.63$ quasar match their predictions and receive $p_{\mathrm{quasar}} = 1.00$ and 0.99–1.00 ($\log \mathcal{R} = 1.6$–1.7 and 1.2). A $z = 2.96$ quasar lies within the predicted ranges but where quasars at that redshift overlap the stellar locus; it is ranked with $p_{\mathrm{quasar}} = 0.44$ and 0.33 ($\log \mathcal{R} = -2.7$ and -1.4). One test quasar at $z = 2.26$ has a Legacy r 2–3 mag fainter than all its other optical bands, a faulty or variable reference measurement; relative to it the colours are impossible for a quasar, and both models decline to rank it (support percentile 0.004). A background star has $\log \mathcal{R} = -24$ and -20 and $p_{\mathrm{quasar}} = 0.00$; its ultraviolet and infrared colours are far from the quasar prediction. The last panel is a background object that both models call a quasar ($p_{\mathrm{quasar}} = 0.97$ and 0.99, $\log \mathcal{R} = 0.5$ at $z_0 = 1.54$): its colours follow the quasar prediction in every band but one, and it is probably an unidentified quasar. Legacy and AllWISE $W1, W2$ appear as separate points because Legacy fluxes are AB and AllWISE Vega; the zero points differ by about 2.7 and 3.3 mag.

10.6 Probability calibration

We test whether p_{quasar} is a calibrated probability on a natural population: all 1,640 known spectroscopic quasars inside the 49 test cones that pass the faint limit, and 40,000 of the cones' background

objects, weighted by 2.75 to restore their natural share (1.47% quasars). The background sample still holds unidentified quasars, so the observed quasar fraction is bracketed by two labellings: known quasars only (a lower bound), and known quasars plus flagged likely quasars (an upper bound).

We separate two questions. The first is abstention. The scorer declines to rank 1.0% and 1.2% of the known quasars (magnitude-independent and magnitude-dependent model), 53% and 62% of the background objects, and 33% and 40% of the flagged likely quasars. Background objects are set aside almost entirely by the support cut (99%); of the 17 and 20 unranked known quasars, 11 and 15 fail the support cut and 6 and 5 lie outside both models. (With the earlier fixed 4σ outside-both test, 3.5% and 3.8% of the known quasars were not ranked, 85% of them outside both models.) The second is the reliability of the returned probability, among ranked objects only (Fig. 21). Ranked objects at $p < 0.01$ contain no known quasars and 1.9–2.2% flagged ones. In the mid range the prediction now lies between the bounds or slightly above the upper one: for example 0.19–0.48 observed at $p \approx 0.57$ and 0.41–0.75 at $p \approx 0.85$ (magnitude-independent; 0.20–0.54 and 0.29–0.74 magnitude-dependent). At the top, $p > 0.99$ gives 0.67–0.97 observed; most of the gap to the lower bound is labelling: 171 of the 188 background objects with $p > 0.97$ (magnitude-dependent model, earlier prior and test) are flagged as likely quasars, and the upper bound lies close to the prediction (`docs/pquasar_diagnostics`). Compared with the earlier abundance prior the mid-range points have moved below the diagonal, because the luminosity-function prior raises p_{quasar} for faint objects. That is the regime where these labels fail: the upper bound counts a background object as a quasar only if it is a Quaia member (Gaia $G < 20$) or has WISE AGN colours, and a quasar at $r \gtrsim 22.5$ usually has neither. The test therefore cannot measure the calibration of faint p_{quasar}, and the excess over the upper bound in the mid range is an upper limit on the overconfidence, not a measurement of it. A two-parameter map of the log-odds, fitted on half the cones, is driven by the large $p \approx 0$ population and pushes mid-range values far below both bounds ($p = 0.5 \to 0.21$–0.23), so it is not applied. p_{quasar} is approximately calibrated for ranked objects at the magnitudes where the labels are complete; an object the scorer declines to rank should be treated as not a quasar candidate, which loses 1.0–1.2% of real quasars in this sample.

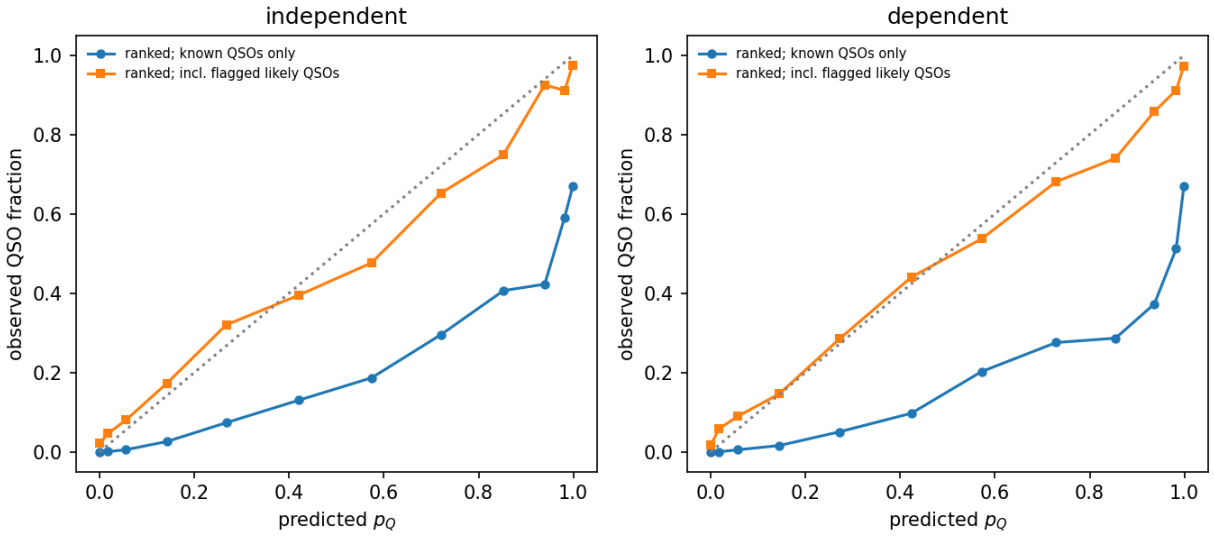


Figure 21: Reliability of p_{quasar} among ranked objects in the 49 test cones (natural population, weighted), for both models: observed quasar fraction against mean predicted p_{quasar} in bins, counting known quasars only (blue, lower bound) or also flagged likely quasars (orange, upper bound). Dotted: perfect calibration. Abstention rates are given in the text.

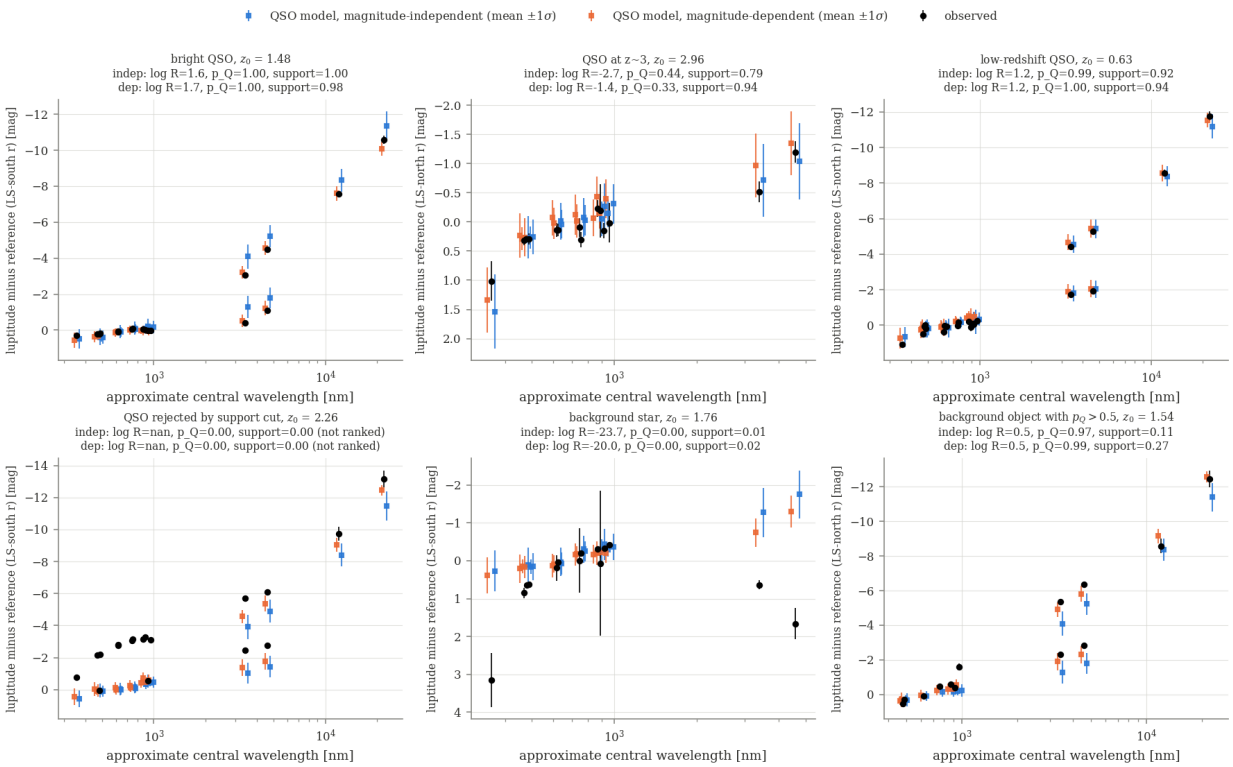


Figure 22: Worked examples: observed luptitude minus reference-band luptitude (black) and each model's conditional prediction at the object's redshift (mean $\pm 1\sigma$; blue magnitude-independent, orange magnitude-dependent). Horizontal positions are approximate central wavelengths, for display only. Titles give $\log \mathcal{R}$, p_{quasar} and the support percentile from both models. Object identifiers are in `docs/method_unified/examples.json`.

11 Blend mode: a known quasar and an unresolved companion

Question. A spectroscopic quasar at known redshift z_0 is blended with an unresolved companion, and only the combined flux is measured. Is the companion a second quasar at z_0 (QQ) or a star (QS)? Nothing is known about the components separately. The mode uses the same quasar and background models as resolved mode (either promoted bundle), accepts any morphology, and takes the best extended photometry of the whole blend (Legacy model flux, SDSS cModel, PS1 Kron). Quasar pairs at different redshifts, and a negligible companion, are further hypotheses to be added later.

Likelihood. Label the known quasar 1 and the companion 2, with reference-band (Legacy r) luptitudes m_1, m_2. For hypothesis H, with companion population C (quasars at z_0, or the background at the blend's sky position),

$$L_H(\boldsymbol{y}) = \iint p(m_1 \mid Q, z_0)\, p(m_2 \mid C,\ \text{detectable})\, p(\boldsymbol{y} \mid m_1, m_2, H)\, dm_1\, dm_2. \tag{10}$$

The magnitude distributions are the bundle's surface densities normalised in magnitude: $\Sigma_Q(z_0, m)$ for the known quasar, and, for the companion, $\Sigma_Q(z_0, m)$ or $\Sigma_B(m, \ell, b)$ normalised over the detectable range only (below). Fluxes add, so we integrate over the companion's share α of the blend's r flux T ($F_1 = (1-\alpha)T$, $F_2 = \alpha T$), with T fixed at the measured value and the Jacobian of $(m_1, m_2) \to (u, \alpha)$, $J = aT\sqrt{T^2 + 4s^2}/\big(\sqrt{F_1^2 + 4s^2}\sqrt{F_2^2 + 4s^2}\big)$, on 48 points uniform in $\operatorname{logit}\alpha$. At each α, each component's mixture is conditioned on its own reference luptitude; for every pair of Gaussian components the blend luptitude in each other band is linearised in the component luptitudes: mean $\operatorname{lup}(F_{1j} + F_{2j})$ and covariance $W_1 C_1 W_1 + W_2 C_2 W_2$ plus the measurement covariance, with $W_i = \operatorname{diag}\big(\sqrt{F_{ij}^2 + 4s_j^2}/\sqrt{(F_{1j} + F_{2j})^2 + 4s_j^2}\big)$ the exact first-order derivative. The instrumental covariance of the native bands enters once, as it belongs to the single blend measurement. The output is $\log L_{\mathrm{QQ}}$, $\log L_{\mathrm{QS}}$, the Bayes factor, $P(\mathrm{QQ})$ for user-supplied prior odds, and the posterior companion share.

Base rules. Extinction is corrected on the blend, since both objects lie behind the same dust. Both components must carry at least 10 times the blend's Legacy r flux error, so the blend needs $S/N \geq 20$ in Legacy r; fainter companions are excluded in this base version. The companion's magnitude prior is therefore conditioned on the companion being detectable: the result is $P(\mathrm{QQ} \mid$ blend, detectable companion), and how often a detectable companion is a quasar rather than a star belongs to the prior odds. (Without this conditioning, the much larger share of stars below the limit penalises QS for companions the mode excludes; in the validation that produced false quasar pairs for faint stars.) No catch-all or support cut is applied, and the known quasar's magnitude prior is Σ_Q, the whole quasar population (Section 9.1); a spectroscopic primary is in practice drawn from the brighter, targeted part, which is an assumption of the mode.

Accuracy of the linearisation. On 79 synthetic blends restricted to Legacy $g, r, z, W1, W2$, the Bayes factor agrees with a brute-force Monte Carlo of the same integral (20,000 flux-space draws per α) in sign for all 79 blends, with median $|\Delta \log \mathrm{BF}| = 0.27$ where $|\log \mathrm{BF}| < 10$. Large differences (15 blends above 5 nat) occur only where QS is clearly preferred: there the Monte Carlo cannot sample the very small QQ likelihood and returns far lower values (for example -99 against -8.8), and the linearisation is the more conservative.

Validation on synthetic blends of held-out objects. We sum the extinction-corrected fluxes (and variances) of real test objects: 1,200 QQ blends of two test quasars in the same redshift slice and 1,200 QS blends of a test quasar and a test-cone background object (likely quasars removed), each component with Legacy r $S/N \geq 10$, both classes placed at random test-cone positions. 99% are scorable.

	current	current_magindep
AUC (QQ against QS, log BF)	0.947	0.927
AUC, companion share $< 10\%$	0.77	0.67
AUC, companion share 10–25%	0.88	0.85
AUC, companion share 25–50%	0.96	0.94
AUC, companion share $> 50\%$	≥ 0.995	≥ 0.994
QQ blends with $P(\mathrm{QQ}) > 0.9$ / < 0.1	75% / 0.6%	63% / 0.6%
QS blends with $P(\mathrm{QQ}) < 0.1$ / > 0.9	67% / 7.2%	63% / 6.9%
median error of companion share, QQ / QS	0.09 / 0.06	0.09 / 0.07

At prior odds 1 on this balanced sample, $P(\mathrm{QQ})$ is close to calibrated (Fig. 23); it is mildly overconfident towards QQ at the top (blends given 0.8–0.95 are QQ in 73–78%), part of which is unidentified quasars among the "stars". With the luminosity-function abundance prior (Section 9.1) the AUC is 0.003–0.009 lower than with the earlier prior, more true QQ blends reach $P(\mathrm{QQ}) > 0.9$ (75% against 71%, and 63% against 60%), and more QS blends do (7.2% against 5.9%, and 6.9% against 5.3%): the prior now makes faint quasar companions more plausible, for real and for star companions alike. The mode separates the hypotheses well whenever the companion contributes a quarter or more of the light, and poorly when it contributes less than a tenth, where the blend looks like the known quasar alone. The magnitude-dependent model separates slightly better, mostly for faint companions. Scoring takes about 0.5 s per blend.

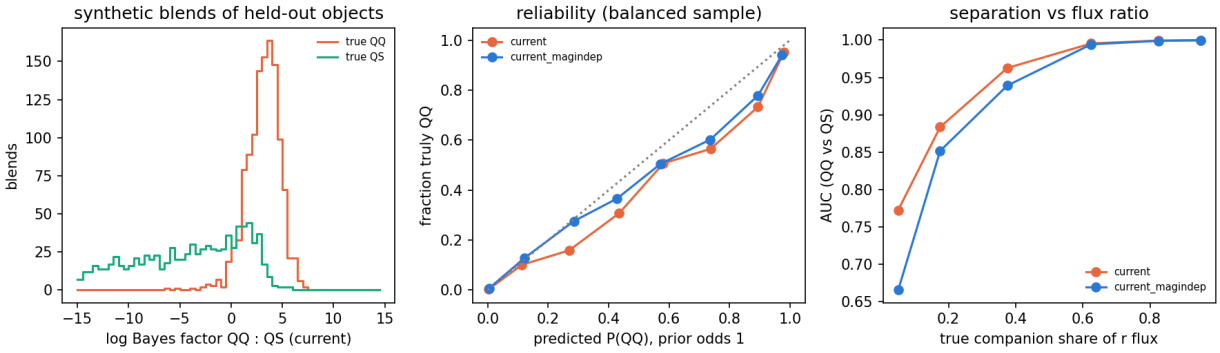


Figure 23: Blend mode on synthetic blends of held-out objects. Left: Bayes factor QQ:QS for true QQ and true QS blends (`current`). Middle: reliability of $P(\mathrm{QQ})$ at prior odds 1 on the balanced sample, both models. Right: AUC against the companion's true share of the r flux.

11.1 What blends look like, and where the hypotheses separate

All illustrations use real held-out objects with Legacy r $S/N \geq 50$, so each component's colours are well measured (`scripts/blend_illustrations.py`; numbers in `blend_illustrations.json`). Background objects are split into four strata of dereddened Legacy $g - r$: < 0.45 (blue), 0.45–0.9 (G–K), 0.9–1.3 (K–M) and > 1.3 (M). These are colour strata of the cleaned background sample, not a physical classification.

Blend colours. Figure 24 sums the fluxes of real component pairs without noise. A QQ blend at the same redshift has the colours of a single quasar at all shares: two draws from one redshift slice average to another member of the slice. A star moves the blend away from the quasar locus in a direction set by its colour. Every star class lowers $z - W1$, because stars lack the quasar's infrared excess; this is the main signature of blue and G–K companions. K and M stars also redden $g - r$ and $r - z$ (M stars raise $r - z$ by 0.4 mag at a share of 0.3) and lower $W1 - W2$. At a share of 0.1 every QS track lies inside, or for M stars at the edge of, the quasar's 16–84% band in every colour. Figure 25 shows the same mixing as tracks in colour–colour planes: from the median quasar at z_0

to the median star of each class. Blue and G–K tracks run almost straight down in $z - W1$ and stay inside the quasar cloud until a share of about 0.3; the M track leaves the cloud sooner, in $r - z$.

Where QQ and QS separate. For the maps we rescale both components of each pair, by one factor per component in all bands, so that the blend has a chosen Legacy r and the companion a chosen share α of the r flux. The noise is sky-limited: the blend variance in each band is the larger of the primary's variance (the depth at its position) and the variance of the rescaled sum, with fresh Gaussian noise making up the difference. QQ companions are test quasars with $|\Delta z| < 0.02$. Each blend is scored at the sky position of a random test-cone object, so position carries no class information. The latitude distributions of the four strata are similar (median $|b|$ 30–32°). Each QS cell is compared with the QQ cell at the same minor share $\min(\alpha, 1 - \alpha)$, since QQ is symmetric. Cells hold 20 blends per class, so a single AUC is uncertain by about ±0.07; the trends across neighbouring cells are the robust result. Scores use 24 points in α.

At $r = 19.5$ (Fig. 26) the companion's share decides the result:

- share ≥ 0.5: AUC ≥ 0.96 in every class and redshift bin;
- share 0.3: median AUC 0.89 (blue), 0.91 (G–K), 0.94 (K–M), 1.00 (M);
- share 0.2: median AUC 0.78, 0.78, 0.88, 0.96;
- share ≤ 0.1: median AUC 0.57–0.74 for blue to K–M companions, close to chance; M stars stay separable down to 0.1 (median 0.87) and partly to 0.05 (0.82).

The redshift dependence is mostly within the cell noise; M-star companions at a share of 0.1 separate better above $z \approx 2.3$ (AUC 0.92–0.95) than at $z < 0.8$ (0.70). QQ blends have a median $\log \mathrm{BF}$ of 1.5–4 in favour of QQ in every cell; between 15% and 80% of them reach $P(\mathrm{QQ}) > 0.9$; the evidence for QQ is weakest in the lowest redshift bin.

With magnitude (Fig. 27, all redshifts) the separation for shares ≥ 0.3 holds to $r = 21.5$. At shares 0.1–0.2 the AUC for K–M and M companions falls by up to 0.2 from $r = 18.5$ to 21.5; for blue and G–K companions it is low at all magnitudes. At $r = 21.5$ the QQ blends' median $\log \mathrm{BF}$ is 0.7–2.1, still in favour of QQ; with the earlier spectroscopic-count prior it was about 0, because that prior had too few faint quasar companions.

Caveats. The maps are controlled experiments, not a census of real blends. The noise is diagonal and sky-limited; real deblending and model-flux errors are not modelled. Rescaling keeps each component's colours but does not change survey detection or the measurement-error model to the new flux. Background objects may include a few unidentified quasars, which lowers the AUC slightly at large shares.

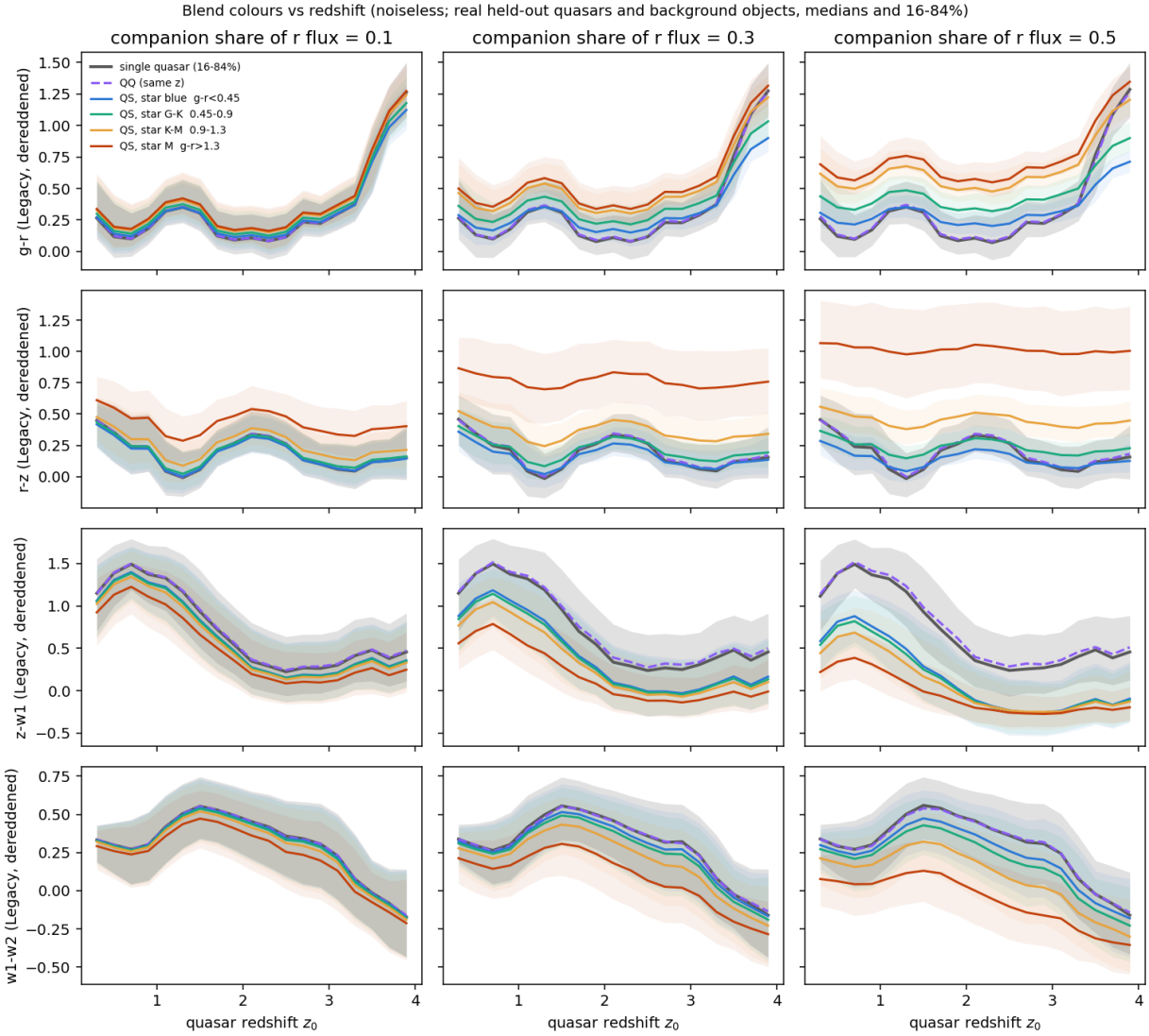


Figure 24: Blend colours (Legacy, dereddened, noiseless) against quasar redshift for companion shares of the r flux of 0.1, 0.3 and 0.5. Grey: single quasars (median and 16–84%). Dashed purple: QQ blends at the same redshift. Coloured: QS blends for the four $g - r$ strata of background objects (median, 16–84% shaded).

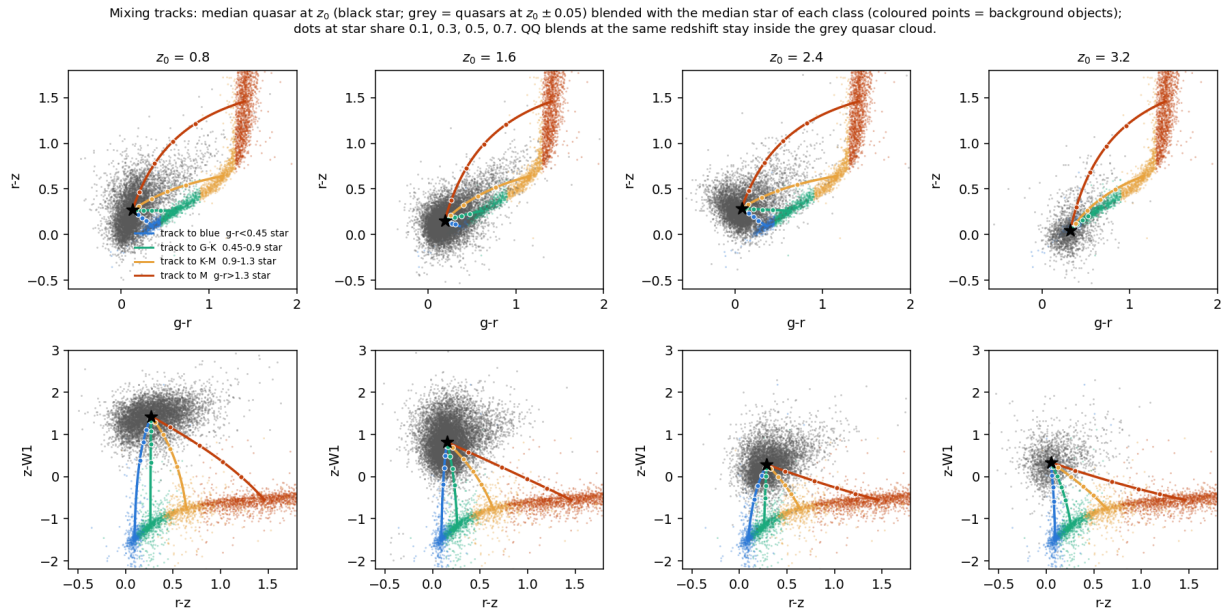


Figure 25: Mixing tracks in $g-r$ against $r-z$ (top) and $r-z$ against $z-W1$ (bottom) at four redshifts. Grey: quasars at $z_0 \pm 0.05$; black star: their median. Coloured points: background objects by $g-r$ stratum. Lines: blend of the median quasar with the median star of each stratum as the star's share of the r flux goes from 0 to 1; dots at shares 0.1, 0.3, 0.5 and 0.7.

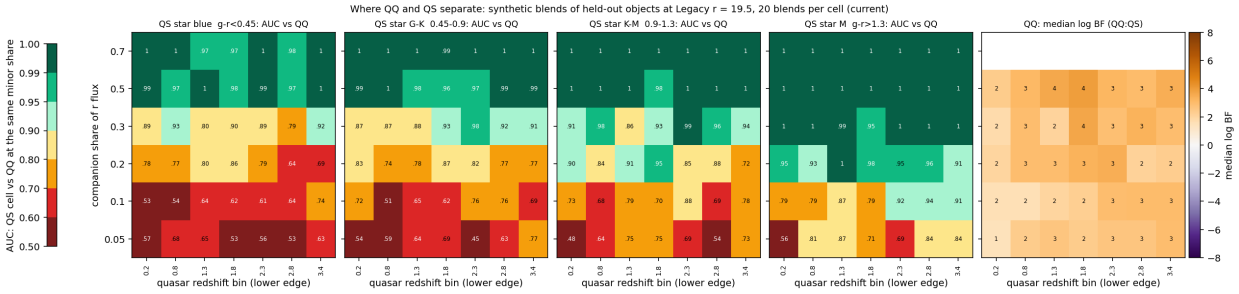


Figure 26: Where QQ and QS separate, at blend Legacy $r = 19.5$. First four panels: AUC of $\log \mathrm{BF}$ between QS blends with a companion of the given $g-r$ stratum and QQ blends at the same minor share, by quasar redshift and companion share of the r flux (20 blends per class per cell, `current`). Last panel: median $\log \mathrm{BF}$ of the QQ blends (minor shares up to 0.5).

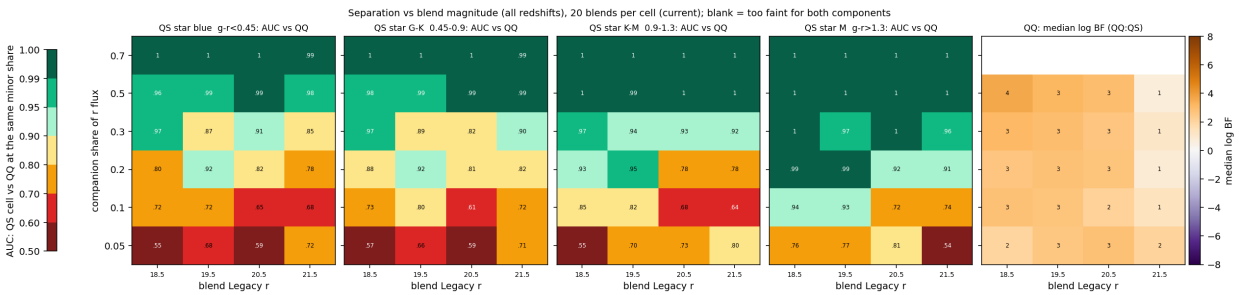


Figure 27: As Fig. 26, against blend Legacy r, with quasar redshifts drawn from the whole test sample.

12 Sources without Legacy r

Why. Every population prior is defined in Legacy r: the quasar abundance $\Sigma_Q(z, r)$, the background abundance $\Sigma_B(r, \ell, b)$ and the magnitude bins of the background colour model. A source without usable Legacy r therefore had no score. Such sources are common outside the Legacy footprint: of 597,964 Quaia quasars ($G < 20$; Storey-Fisher et al., 2024) at $|b| \geq 25°$, 115,979 have no Legacy DR9 source within $1''$ (mostly at Dec $< 0°$, 42k below $-30°$) and 37,267 have one with masked r (`docs/fallback_reference/quaia_no_legacy_r.json`).

Method. Legacy r stays the population reference and is treated as unknown. With m the noise-free Legacy South r luptitude (the latent `legacy:r`), each scorer intensity H is

$$\Lambda_H = \int \lambda_H(m)\, dm, \qquad \lambda_H(m) = \Sigma_H(m)\, p_H(\boldsymbol{y} \mid m), \tag{11}$$

where $\boldsymbol{y}$ are all observed bands. $\lambda_H(m)$ is exactly what the scorer returns for the same source with a noise-free Legacy r measurement equal to m, so the integral reuses the scorer: the source is replicated on 41 values of m and the intensities are integrated by the trapezoid rule. In blend mode m is the blend's Legacy r, and the blend likelihood is integrated in the same way. No model is refitted: the joint colour models were fitted to objects observed in Legacy and the other surveys at once, so $p_H(\boldsymbol{y} \mid m)$ carries the learned relation between Legacy r and the other bands. A *fallback band* (PS1 r, else NSC r, else SkyMapper r) at $S/N \geq 10$ (≥ 20 for blends, with the detectability rule applied to it) makes a source eligible and centres the integral. The outside-both-models test and the support cut are evaluated at the posterior mode of m. This is the default; each call that uses it raises a warning, and affected sources carry the flag `legacy_r_marginalised:<band>`.

The integration window is $\pm(1.2\,\mathrm{mag} + 5\sigma)$ about the fallback magnitude, σ its error. It is a support restriction. Measured on the training samples (both bands $S/N > 30$), the fallback-minus-Legacy-South r scatter (NMAD) is 0.21 mag for quasars in PS1 (mostly variability between the two epochs; 0.04 in NSC, which uses the same DECam imaging, and 0.13 in SkyMapper) and 0.03–0.09 mag for stars; offsets beyond 1 mag occur for 0.4% of quasars in PS1, beyond 1.5 mag for 0.03%. Far outside this range the models' $p_H(\boldsymbol{y} \mid m)$ is an extrapolation: at faint m the components were never constrained by fallback-band detections, and the steeply rising $\Sigma_B(m)$ turns their tails into spurious mass. On the hardest test objects p_{quasar} changed by a median 0.003 between half-widths of 0.8, 1.2 and 1.6 mag, but by up to 0.2 between 1.6 and 3 mag. The models reproduce the PS1 and NSC relations for stars, but give quasars a conditional scatter of 0.25 mag in PS1, 0.14–0.28 in NSC and 0.7–1.0 in SkyMapper; only 2,349 training quasars have both SkyMapper and Legacy r.

Validation. Held-out objects that do have Legacy r (test roles, outside every development exclusion, fallback band at $S/N \geq 10$) are scored twice with the same remaining bands: with their true Legacy r, and with Legacy r removed and integrated over. The PS1 test removes all Legacy bands; the NSC test also removes PS1 and SDSS (the deep south); the SkyMapper test also removes NSC. Table 2 gives the results; 3,000 quasars and 3,000 background objects per resolved-mode test (all 1,061 qualifying quasars for SkyMapper), and 400 QQ and 400 QS synthetic blends per blend-mode test.

Findings. Ranking is nearly unchanged with PS1 and NSC r: AUC losses are 0.002 or less in resolved mode, and there is no loss in blend mode. Without Legacy r, p_{quasar} is lower: 3–6 percentage points fewer quasars reach 0.5, while the fraction of background objects doing so changes by at most 0.05 points, so the fallback is conservative rather than noisier. Blend Bayes factors shift towards QQ by 0.4–0.5 nat, and keep their sign for 94–98% of blends. NSC r recovers Legacy r best (NMAD 0.08 mag), as expected from the shared imaging. SkyMapper r is weak: Legacy r is recovered with a bias of 0.15–0.21 mag, and in blend mode the sign of $\log \mathrm{BF}$ changes for a fifth of blends. SkyMapper-only sources should be treated as indicative.

Fallback	N_Q	$\mathrm{AUC}_{\mathrm{true}\,r}$	Resolved mode AUC loss	recall loss (pp)	Δbkg (pp)	Δr (NMAD) (mag)	Blend mode AUC loss	$\Delta \log \mathrm{BF}$ (nat)	sign (%)
PS1 r	3,000	0.9940 / 0.9942	0.0020 / 0.0016	5.5 / 5.0	−0.23 / −0.12	+0.03 / +0.03 (0.22 / 0.21)	−0.008 / −0.003	+0.4 / +0.4	97 / 98
NSC r	3,000	0.9786 / 0.9798	0.0009 / 0.0008	3.7 / 2.9	+0.00 / +0.05	+0.02 / +0.02 (0.08 / 0.08)	−0.005 / −0.004	+0.5 / +0.5	94 / 94
SkyMapper r	1,061	0.9887 / 0.9894	0.0049 / 0.0054	2.4 / 1.2	+0.08 / +0.25	+0.21 / +0.15 (0.35 / 0.28)	+0.012 / +0.014	+1.6 / +1.3	81 / 79

Table 2: Scoring without Legacy r, against the same objects scored with their true Legacy r and the same remaining bands (first number magnitude-independent model, second magnitude-dependent). $\mathrm{AUC}_{\mathrm{true}\,r}$: with the true Legacy r. Recall loss: drop in the fraction of quasars with $p_{\mathrm{quasar}} > 0.5$. Δbkg: change in the fraction of background objects with $p_{\mathrm{quasar}} > 0.5$. Δr: posterior-mean minus true Legacy r of quasars, median (NMAD). Blend mode: AUC loss of log BF, median shift of log BF, and the fraction of blends whose log BF has the same sign as with the true Legacy r.

12.1 Point sources outside Legacy

Resolved mode scores only point sources, and outside the Legacy footprint there is no Legacy `type`. `qso_pcolor.point_source` supplies a replacement: where a source has a PS1 detection, its PS1-PSC random-forest score (Tachibana & Miller, 2018) with the authors' figure-of-merit threshold `ps_score` ≥ 0.83; otherwise the Gaia DR3 corrected BP/RP flux excess C^*, a point source if $|C^*| < 3\,\sigma_{C^*}(G)$ (Table 2 and Eq. 18 of Riello et al., 2021); otherwise no test, and the source is not scored. Against Legacy `type` inside the footprint (106,519 field sources in 40 cones of 0.15°; 20,000 Quaia quasars), the combined test calls 97–99% of Legacy point sources at $r < 20$ point, 88–95% at $20 < r < 21$ and 57–75% at $21 < r < 22$, and fewer than 1% of extended sources at any magnitude; it calls 95–97% of Legacy-point quasars at $18 < r < 20.5$ point, but only 80–91% at $r < 18$. Gaia alone, which is all there is south of Dec $= -30°$, is weaker: it calls 60% ($r < 17$) to 83% ($17 < r < 18$) of Legacy-point quasars point — a C^* cut also removes variable sources, as Riello et al. (2021) note — and passes 10–15% of extended sources at $r = 20$–21. PS1-PSC is therefore used first. The background model still describes Legacy `type=PSF` sources; the small mismatch between the two definitions of a point source is not modelled (`docs/point_source/validation.json`).

13 Limitations and open items

- **Probability calibration** is approximate and untested at faint magnitudes: among ranked objects p_{quasar} lies between or slightly above the label bounds in the mid range, and the labels miss most quasars at $r \gtrsim 22.5$; 1.0–1.2% of real quasars are not ranked (Section 10.6); the reliability of p_{same} for a narrow redshift window is not tested.
- **Background purity.** Faint unidentified quasars not detected by WISE remain in the background training sample and counts. The test background sets carry the same contamination, which lowers the measured AUC slightly.
- **Faint limit.** Sources below Legacy r $S/N = 10$ are not scored. Sources without Legacy r are scored through a fallback band, less precisely, and weakly where only SkyMapper r is available (Section 12).
- **Galactic latitude.** All training samples are at $|b| \geq 25°$, and both modes refuse sources below it (Section 4); lowering the limit needs training cones at lower latitude.
- **Unranked quasars.** 0.6% of test quasars above the faint limit are not ranked, mostly at the support cut. The earlier 3% came from the fixed 4σ outside-both test, now replaced (Section 9.2). Variability between surveys observed years apart is not modelled; it makes some quasars' photometry inconsistent across surveys (`docs/pquasar_diagnostics`).
- **Selection.** Test quasars share the training selection. Our tests cannot measure the benefit of the magnitude-independent model for quasars outside the DESI and SDSS selection, so the choice between the two models (Section 6.2) rests partly on that untested risk.

- **Quasar abundance** is the eBOSS luminosity function, extrapolated beyond $r \approx 22.3$ and below $z = 0.68$, with an interpolated redshift shape inside its $\Delta z = 1$ bins (Section 9.1).
- **Sparse surveys.** NSC VR and VISTA Y extinction coefficients are approximate, and SkyMapper u, v and VHS have little training support.
- **Background in the faintest bins** ($r > 21.5$) is broader than the data in $W1 - W2$ and slightly narrower in $r - z$ against $z - W1$ (Fig. 8); these panels hold few WISE-detected stars.
- **Out of scope.** Blends below the separation floor, and a physical-pair clustering prior, are deferred.

A The window factorisation

Write $\Lambda_Q = \int \lambda_Q(z)\, dz$ for the total quasar intensity, and let the declared window W around z_0 have effective width $\Delta z_{\rm eff} = \int_W k(z)\, dz$ for its kernel k. The posterior that the companion is a quasar in the window is

$$p_{\rm same} = \frac{\int_W k(z)\lambda_Q(z)\, dz}{\Lambda_Q + \lambda_B + \lambda_U} = \Delta z_{\rm eff}\, \frac{\langle \lambda_Q \rangle_W}{\Lambda_Q + \lambda_B + \lambda_U} \equiv \Delta z_{\rm eff}\, \mathcal{R}, \tag{12}$$

with $\langle \lambda_Q \rangle_W$ the kernel-weighted mean over the window. The denominator contains no window, because Λ_Q includes both same-redshift and other-redshift quasars. The scorer integrates the window on dense nodes at the support boundaries rather than on the coarse redshift grid. For a window much narrower than the photometric redshift width, $\langle \lambda_Q \rangle_W = \lambda_Q(z_0)$, which is Eq. (3).

B The constrained M step

For a component with mean $(\mu_m, \boldsymbol{\mu}_c)$ and covariance $\mathrm{diag}(\sigma_m^2, V_c)$, where μ_m and σ_m^2 are fixed, the expected complete-data log likelihood separates into a magnitude term, constant in the free parameters, and a colour term. Maximising the colour term with the MAP prior on V_c gives the usual updates applied to the colour block alone. The prior on the full covariance differs from the prior on V_c by a constant. The constrained update is therefore exact.

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